OECD Economic Surveys

Luxembourg

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OECD ECONOMIC SURVEYS

2000-2001

LUXEMBOURG

OECD

ORGANISATION FOR ECONOMIC CO-OPERATION AND DEVELOPMENT

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BASIC STATISTICS OF LUXEMBOURG (1999)

THE LAND

Area (km^2)	2 586	Major city, thousand inhabitants:	
Agricultural area (km^2)	1 274	Luxembourg, 1.3.1991	75.8
Woodland (km^2)	886		

THE PEOPLE

Population (thousands), 1.1.99	435.7	Employment (thousands):	
Inhabitants per km^2	168	Total domestic employment	248.3
Net natural increase	1 789	Dependent employees:	231.5
Net migration	4 719	Agriculture	1.6
		Industry and construction	58.4
		Services	171.5
		Employers, self-employed persons and domestic help	16.8

THE PRODUCTION

Gross domestic product (billion LF)	731.8	Gross domestic product by origin,	
Gross domestic product per head (US$)	44 691	at basic prices (per cent):	
Gross fixed investment:		Agriculture	0.7
Per cent of GDP	22.4	Industry and energy	14.6
Per head (US$)	10 012	Construction	6.3
		Other	78.4

THE GOVERNMENT

Per cent of GDP:		Composition of the Chamber (number of seats):	
General government consumption	17.7	Christian-socials	19
General government current revenue	47.1	Socialists	13
General government gross debt	6.0	Democrats	15
		Others	13
		Total	60
		Last election: 12.6.1999	

THE CURRENCY

Irrevocable conversion rate	40.3399	Currency units of euro per US$, average of daily figures:	
		Year 2000	1.1108
		December 2000	1.0851

N*ote*: An international comparison of certain basic statistics is given in an annex table.

This Survey is based on the Secretariat's study prepared for the annual review of Luxembourg by the Economic and Development Review Committee on 18 December 2000.

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After revisions in the light of discussions during the review, final approval of the Survey for publication was given by the Committee on 17 January 2001.

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The previous Survey of Luxembourg was issued in January 1999.

Assessment and recommendations

High growth in output and employment continues

Luxembourg's long-running economic boom continues apace, with growth in real GDP having risen to 7½ per cent in 1999, significantly above the long-term average (5½ per cent since 1985). This strengthening in the economic expansion, which occurred in the context of slowing growth in the EU, was led by the private service sectors – notably transport and communications and financial services – where sustained dynamism has weakened the influence of the international business cycle on the economy. Buoyed by the strength in economic activity, domestic employment growth rose to a record 5 per cent in 1999. As usual, cross-border workers took most of the new jobs. Unemployment continued to edge down to less than 3 per cent. Economic activity appears to have strengthened further this year. Again, growth has been very strong in the private services sectors, notably finance. In addition, there has been a sharp turnaround in the manufacturing sector owing to more favourable international market conditions, especially in the important steel industry. Domestic employment growth has increased further to around 5½ per cent (year-on-year) in recent months. These developments point to economic activity expanding at around 8 per cent in 2000.

But inflation has risen, boosting index-linked wage increases

Inflation has picked up markedly, to around 3½ per cent by late 2000, mainly reflecting the rise in oil prices. Nevertheless, underlying inflation also rose to around 2 per cent by the second quarter of 2000, and has continued at this rate since then. As a high proportion of goods and services consumed in Luxembourg is imported, a significant part of this rise in inflation is attributable to price developments in neighbouring countries. But high wage increases in Luxembourg since late 1999 have also played a role. They have

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been boosted by indexation arrangements, which automatically raise wages (and social security benefits) whenever cumulative increases in the CPI reach 2½ per cent. In the wake of the oil price shock and the substantial depreciation of the euro, an indexation threshold was breached in August 1999, two and a half years after the previous occasion on which an indexation threshold had been breached, and again in June 2000. This has boosted hourly wage increases in Luxembourg from among the lowest in Europe over 1996-98 to among the highest. While the 1975 law on wage indexation provides a means of breaking a vicious circle of price and wage increases by suspending automatic indexation in difficult economic circumstances, this provision is rarely used; it has only been invoked once, at the time of the devaluation of the Luxembourg franc in the early 1980s. In view of the exceptional circumstances required to suspend indexation, it would be preferable to exclude energy prices from the relevant index (as in Belgium) so as to reduce the risk that the increase in energy prices sets off a wage-price spiral.

Tax cuts will reduce the government budget surplus

The general government budget surplus rose by 0.7 percentage point of GDP in 1999 to 4.4 per cent. This increase was mainly attributable to a decline in expenditures as a share of GDP, reflecting ceilings on public employment growth and limited wage increases in the public sector. The budget surplus is estimated to remain unchanged in 2000, despite an increase in government investment expenditure, owing to a further decline (as a share of GDP) in pension outlays. Over the projection period, the budget surplus (on a cash basis) is projected to decline markedly to 2½ per cent of GDP, reflecting the implementation of tax cuts for both households and businesses. These tax cuts should have beneficial effects on labour market participation and on the attractiveness of Luxembourg as an investment destination. Although these tax cuts will also stimulate demand at a time when the economy is already growing strongly, they are unlikely to provoke overheating in view of the elastic labour supply and the high import content of consumer expenditure. The tax cuts will not alter the fact that Luxembourg has strong public finances; general government gross debt was only 6 per cent of GDP in 1999; and net financial assets amounted to about

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30 per cent of GDP. The high level of net financial assets needs to be seen in the light of the large and rapidly growing future (PAYG) pension commitments associated with the high rate of employment growth since the mid-1980s. If this growth rate were to slow significantly and no further measures were taken, very large increases in contribution rates/taxes would be required to meet maturing pension commitments. In view of this risk, the authorities have commissioned a report on pension financing which will study the conditions for financing these commitments in the long term and examine whether it would be opportune to make certain modifications, such as placing greater reliance on private pension funds. While moving in this direction would entail transitional costs, it would enhance the long-term sustainability of fiscal policy in Luxembourg.

Growth and inflation should slow from a peak in 2000

Although fiscal policy is set to ease, the effects of this on growth are likely to be outweighed by slowing growth in export markets and an assumed moderate tightening in monetary conditions in the euro area. These factors underpin a projected decline in the growth rate to around 5½ per cent by 2002. Inflation is likely to peak in 2000 at 3 per cent or so and, with the passing of the oil price shock and euro depreciation, to fall back to around 2 per cent by 2002. The main risk to these projections is that the energy-price shock leads to a serious wage-price spiral owing to indexation arrangements.

Further structural reforms would raise economic efficiency and better prepare the economy to withstand adverse shocks

Extremely rapid expansion in the finance sector underpins to a considerable extent the sustained high economic growth rates in Luxembourg since the mid-1980s. The increase in economic growth compared with earlier periods was achieved through raising growth in productive inputs: multi-factor productivity (MFP) growth declined. While rapid growth in employment has sheltered residents from the large increases in unemployment that occurred in other European countries with similar labour market institutions, it has not resulted in a significant increase in the (national) employment ratio, mainly owing to inappropriate incentives. These developments suggest that the need for structural reform to enhance economic efficiency remains as pressing as ever, despite the high economic growth and low unemployment experienced since the mid-1980s. Moreover,

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the authorities would do well to reform some of the more employment-unfriendly labour market institutions, such as high benefit replacement rates and long benefit duration (taking into account social assistance), before they have a chance to interact with a sustained decline in employment growth, which would be likely to result in a large increase in structural unemployment.

Additional reforms are required to raise the employment ratios for older workers…

The 1999 law on the National Action Plan (NAP) to Promote Employment contained some measures to increase the employment ratio for older workers, which is low by international comparison, namely by creating the possibility of a progressive early retirement (as from age 57) or of a progressive shift to part-time employment (for workers aged over 49). The government also plans to address concerns about the profitability of employing older workers by offering them training in the use of information technology and, when they are no longer as productive as before, by providing financial assistance to employers and/or, if wages are cut, to employees. In addition, the government, in consultation with the social partners, intends to reform the law on disability pensions. If effective, this could be particularly potent in raising the employment ratio for older workers as they are over-represented among disability pensioners. Access to early retirement pensions should also be restricted and/or the pensions should be made less attractive to discourage older workers from taking an early retirement. Pre-retirement pensions (*préretraite*), which finance retirement of older workers at ages as low as 54 in sectors experiencing difficulties, should be phased out. They encourage premature retirement and impose a tax on the rest of the economy, slowing the transfer of resources to more productive sectors. At the same time, the qualification age for early retirement pension (*retraite anticipée*), which can be as low as 57, should be raised and/or the pension made less generous to discourage early retirement. One option for reducing the generosity of these pensions would be to reduce them actuarially relative to a pension taken at the official retirement age (65). This could be complemented by an actuarially based increase in pensions taken after the official retirement age. Increasing the employment ratio for older workers would also reduce the budgetary cost of financing pensions.

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... and for women

The government has also taken measures to increase female participation in the labour force, which is also low by international comparison. It has introduced professional training addressed exclusively to women and encourages females to choose scientific and technical subjects in their education. A legal basis has been introduced for positive discrimination in private businesses and arrangements have been made to pay subsidies to employers hiring members of the underrepresented sex in certain professions. The government has also started to take steps to lower the barriers to female participation in the labour force posed by the difficulty of reconciling family and work life. It has introduced parental and family leave as well as subsidies to private businesses that re-organise working arrangements to facilitate the reconciliation of family and working lives. In addition, the government plans a major expansion in day-care places. Moreover, flexibility of working time under part-time contracts has been increased, which should expand employment opportunities for women as they hold most part-time jobs; work time flexibility has also been increased for full-time contracts. Nevertheless, substantial regulatory barriers continue to dissuade employers from offering part-time jobs, depriving women of an important means of reconciling family and working life. The government, in consultation with the social partners, is planning to reform the law on part-time employment to remove some of these barriers. Such a reform should facilitate the development of part-time employment, thereby contributing to a narrowing of Luxembourg's large gender gap in the employment ratio as well as lifting the overall employment ratio. The authorities should also consider moving from the household to the individual as the basic unit for taxation so as to provide women with greater incentives to participate in the labour force.

Relaxation of strict employment protection regulation could help certain job seekers to obtain employment

Luxembourg has strict employment protection regulation (EPR), notably concerning dismissals and the use of fixed-term (including temporary) employment contracts. Some relaxation in EPR would probably help job seekers among older working age persons, prime-age women and youths to obtain employment by giving employers greater flexibility to adjust employment to adverse demand shocks, making them less reluctant to hire members of these

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groups. This could be particularly important for older job seekers in the event that the measures to discourage older workers from withdrawing prematurely from the labour force are successful. While less strict rules on dismissals and the use of fixed-term contracts would probably increase the number of older workers who are laid-off by their employers, the measures planned to increase the employability of older workers increase the likelihood that such a reform would raise the employment ratio for older workers. Based on the tentative international evidence from a recent OECD study, relaxing EPR would probably also contribute to higher MFP growth by facilitating the reallocation of resources to more productive uses.

Progress should continue to be made in liberalising network industries

As in other EU countries, implementing EU Single Market directives has entailed substantial reforms in major network industries. This has involved separating natural monopoly elements from potentially competitive ones and opening the latter to competition while at the same time maintaining universal service obligations. Luxembourg opened all but the local loop of the telecommunications industry to competition in 1998 and allows competition for about 40 per cent of the electricity market. Legislation has been passed providing for a progressive opening of this market to competition over 2001-05 for business customers. Legislation has also been introduced for opening the gas market to competition. Overall, telecommunications prices have fallen sharply since liberalisation, especially for international calls, with the result that charges in Luxembourg are now amongst the lowest in the OECD. However, failure to open the "last mile" to competition has kept charges for local calls, including Internet usage, relatively high. This might have slowed the spread of Internet usage, countering the influence of high disposable income and of the high proportion of households with a personal computer, which favour the rapid diffusion of Internet usage. In view of the likely importance for productivity growth of removing barriers to the diffusion of information and communications technology (ICT), it is important that the authorities take steps to reduce the costs of Internet usage. The opening of the local loop to competition from January 2001 should have this effect. This will complement the package of reforms – elec-

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tronic signature, consumer protection and protection of personal information – aimed at adapting the legal framework to the needs of electronic commerce.

The financial sector is evolving away from being based on regulatory, tax and other advantages,…

Luxembourg's financial sector is evolving from being, based on regulatory, tax and other advantages towards being based mainly on specialised knowledge and clustering benefits. For example, funds management and securities clearing activities have grown considerably in importance relative to private banking. The authorities have long adopted a pro-active approach to financial market regulation to ensure that it facilitates development in the sector. This has involved identifying and quickly removing any legal barriers to financial market niches. In keeping with this approach, the government recently passed new laws on mortgage banking and fund management, which should facilitate development of these activities. Moreover, as noted above, new laws on electronic commerce have been passed or are being prepared, which should stimulate development in the provision of financial services *via* the Internet.

… putting it in a better position to follow the international trend towards co-operation in tax matters

In view of this ongoing evolution in the financial sector, it is in a better position to follow the international trend towards co-operation in tax matters. This evolution is timely, as agreements have been made to work towards increasing such co-operation. All OECD countries, including Luxembourg, agreed in March 2000 to work towards improving access to bank information for tax purposes. Luxembourg's Parliament has already taken a step in this direction by finalising the ratification of the additional Protocol on tax matters to the European Convention on Mutual Assistance in Criminal Matters in October 2000. This means that signatories to the Convention that have also ratified the additional Protocol will now be able to request information, including bank information, from Luxembourg for the investigation and prosecution of criminal tax matters. At the same time, Parliament also ratified a similar agreement with the United States. Moreover, it is to be noted that the EU Council agreed at the Feira Summit in June 2000 on key principles for a future Directive on the taxation of savings. This Directive, which will apply only to non-resident individuals of another member state of

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the EU, aims to establish after the end of the transition period a system of automatic exchange of information on cross-border interest flows to ensure a minimum effective taxation of savings income in the form of interest payments within the EU. Implementation of the different stages is conditional on equivalent measures being taken in the United States and key third countries (including Switzerland) and the same measures being taken in all relevant dependent or associated territories (including the Channel Islands) of EU members. The agreement allows Luxembourg to operate a withholding tax instead of participating in the information exchange network during the transition period. Once sufficient reassurances have been obtained concerning the application of equivalent or the same measures in the named countries or territories of EU members, the EU Council will decide by unanimity no later than 31 December 2002 on the adoption and implementation of the Directive. The ECOFIN Council at its meeting of 26 November 2000 recalled the conditions governing transition from each stage to the next as set out in its Report to the European Council at Feira and its annex and approved the essential content of the future Directive on the taxation of savings income. Luxembourg is also participating in the OECD project to identify and eliminate harmful tax practices involving geographically mobile financial and other service activities. A number of tax regimes in Luxembourg have been identified as being potentially harmful. These, along with potentially harmful tax regimes in other countries, are to be examined with a view to determining whether they are actually harmful and, in the event that they are, how the harmful features could be removed. As Luxembourg abstained from the initial (1998) Report on harmful tax practices, it reserves the right not to remove such features of its preferential tax regimes.

Summing up

Luxembourg continues to build on a remarkable economic performance, maintaining high growth rates and low unemployment. Nevertheless, multi-factor productivity growth has slowed and labour market institutions expose the country to the risk of a large increase in structural unemployment if the leading industries were to be hit by an adverse shock. Further progress in structural reform would

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enhance multi-factor productivity growth, increase the proportion of the working-age population in employment and better prepare the country to weather adverse shocks. A concern about the short-term outlook is that inflation has picked up markedly to about 3½ per cent (2 per cent in underlying terms), contributing to a sharp acceleration in wage increases through indexation arrangements. Modifying these arrangements to exclude energy prices from the relevant index (as in Belgium) would reduce the risk that the increase in energy prices sets off a wage-price spiral with an associated loss of competitiveness and weakening of growth. Looking further ahead, changes to pension arrangements would be required to contain the budget cost of the public PAYG scheme if employment growth were to slow significantly from the high rates recorded since the mid-1980s. In the more immediate future, international co-operation in tax matters could enhance the ongoing trend of rapid financial sector diversification, reinforcing the pro-active approach of the authorities which ensures that the legal and regulatory framework remains conducive to development of financial market niches. In sum, policies that allow greater adaptability in the economy, especially in the labour market, would increase the likelihood that Luxembourg's impressive economic performance can be sustained for many years to come.

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I. Recent trends and short-term prospects

Economic activity

Growth accelerated in 1999[1]...

Real GDP growth rose to 7½ per cent in 1999, well above the long-term average growth rate (5½ per cent since 1985). Although this may slightly overstate underlying trends because a new system for elderly care (*l'assurance dépendance*) increased public consumption[2] and hence GDP growth, there was nevertheless a broadly based acceleration in economic activity (Table 1). The largest increase in expenditure was in gross fixed investment, even after abstracting from one-off purchases of aircraft, which account for about one-half of the overall growth in fixed investment.[3] The strengthening in investment reflected an improving international economic outlook, high profitability, low interest rates and high levels of capacity utilisation. Growth in private consumption expenditure also strengthened significantly, reflecting an increase in population growth (through higher net immigration) and rising household incomes. In addition, there was a sharp increase in public consumption (8 per cent abstracting from the introduction of the new system for elderly care). The strength in domestic demand resulted in a very large increase in imports, slashing the net foreign contribution to growth.

... led by strong growth in the transport and communications sector

Most of the increase in growth in 1999 was attributable to the transport and communications sector, where real value added leapt by 28 per cent. This increase was attributable to further growth in air transport services (especially airfreight), satellite communications and mobile telephone services. There was also a large increase (19 per cent) in the health and social action category, reflecting the introduction of the new system of elderly care. Growth in the financial services sector, which now accounts for 20 per cent of GDP, accelerated in line with expansion in the overall economy. The increase in real value added in this sector was mainly attributable to a rise in banks' commissions. As in other European countries, the manufacturing sector was a drag on growth (it contracted by 1.4 per cent) owing to the weak international business environment through till the first half of the year.

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Table 1. **Demand and output: recent trends and projections**
Annual percentage changes, 1995 prices

	1997 current prices		1998	1999	2000[1]	2001[1]	2002[1]
	LUF billion	Per cent of GDP					
A. Demand and output							
Private consumption	289.0	46.3	2.3	4.1	3.5	5.5	5.0
Government consumption	107.9	17.3	2.8	12.8	3.9	3.6	3.6
Gross fixed investment	125.8	20.1	1.5	26.6	0.1	3.2	4.0
Final domestic demand	522.8	83.7	2.2	11.4	2.6	4.5	4.4
Stockbuilding[2]	1.8	0.3	0.1	0.1	–0.1	0.0	0.0
Total domestic demand	524.6	84.0	2.3	11.5	2.5	4.5	4.4
Exports of goods and services	685.5	109.8	9.9	7.9	15.0	10.0	8.3
Imports of goods and services	585.6	93.8	8.3	11.2	11.2	9.3	8.0
Foreign balance[2]	100.0	16.0	3.0	–1.9	6.0	2.6	2.0
GDP at constant prices	..	..	5.0	7.5	8.1	6.2	5.5
GDP deflator	..	..	1.5	2.3	2.6	2.8	2.1
GDP at current prices	624.6	100.0	6.6	9.9	10.8	9.2	7.7
B. Memorandum items:							
Private consumption deflator	..	..	1.7	1.4	3.1	2.3	2.1
Industrial production	..	..	4.3	3.1	8.0	6.0	4.0
Total employment[3]	..	..	2.0	2.4	2.5	2.0	1.8
Unemployment rate (per cent)	..	..	3.1	2.9	2.7	2.5	2.3

1. Projections.
2. Contribution to growth of GDP.
3. National concept.
Source: OECD.

Private services underpin high growth and weaken the effects of the international business cycle

Taken together, the financial services- and the transport and communications sectors were the mainstay of growth in 1999, repeating the pattern of recent years (Figure 1). Real value added in these sectors grew by 44 per cent over 1995-99, accounting for almost two-thirds of the total growth in real value added (20.8 per cent); the share of these sectors in total value added rose from 32 per cent to 37 per cent. The rest of the economy grew at an average annual rate of 2.9 per cent over this period, which is more in line with growth rates in other European countries. These high growth sectors also appear to be much less influenced by the European business cycle than the rest of the economy, thereby weakening the influence of the business cycle on the Luxembourg economy (Figure 2). It is this factor which underlies the fact that the Luxembourg economy boomed in 1999

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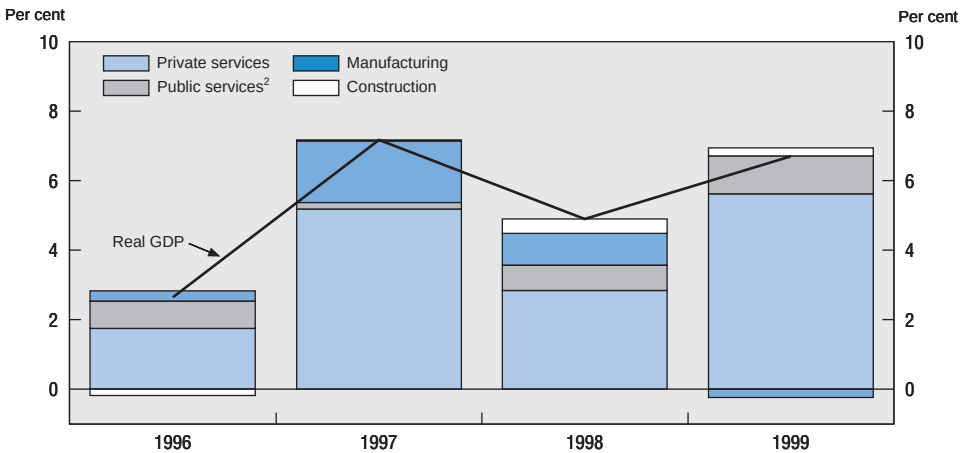
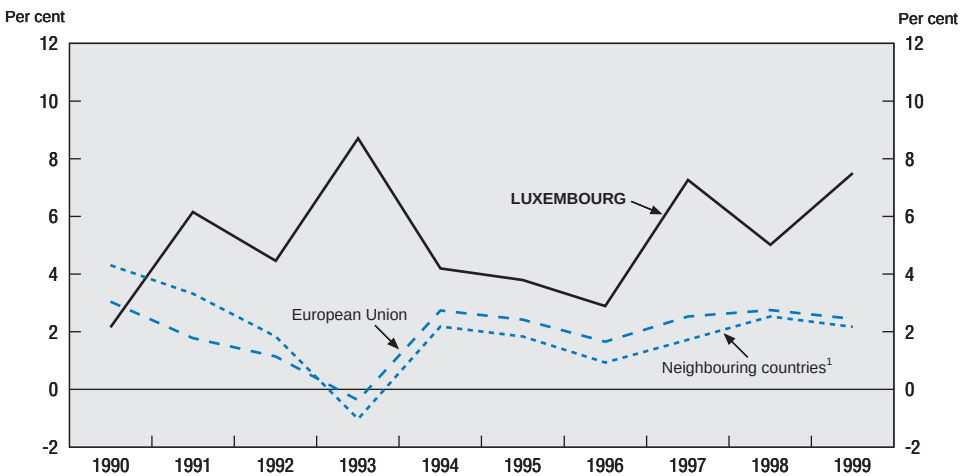
Figure 1. **Contributions to real GDP growth**[1]
As a percentage change of GDP in previous year



1. GDP, 1995 prices.
2. Including public administration, education, health, social services and personnel services.
Source: STATEC.

Figure 2. **Real GDP growth**
Per cent change over previous year



1. Belgium, France and Germany.
Source: OECD, *National Accounts*.

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despite the marked slowdown in growth that occurred in the rest of Europe in the aftermath of the Asian and Russian financial crises.

Business indicators point to even higher growth in 2000

Growth in the services sector has risen further this year, notably in the financial sector. Growth in the financial sector is broad-based, but has been especially rapid in funds management, with growth in assets under management rising to 48 per cent (year-on-year) in the first nine months of 2000, compared with 31 per cent in 1999. Strong growth in the financial sector has contributed to high growth in other service sectors, notably business services. VAT returns show that (nominal) turnover for market-sector services other than financial services, rose by 20 per cent (year-on-year) in the first ten months of the year. Growth in the manufacturing sector has also strengthened, especially through to the first half of 2000 and in the (large) steel industry, reflecting the international upswing. Industrial production rose by 5 per cent (year-on-year) in the first ten months of the year, compared with an increase of 1.5 per cent in 1999. Although capacity utilisation in manufacturing rose to 88 per cent by the first quarter of 2000, the highest since 1992, and has since remained around this level, few businesses reported feeling constrained by a lack of capacity (4 per cent) or by a lack of labour (1 per cent) in the third quarter. Business conditions are also very favourable in the construction industry, with turnover up by 8 per cent (year-on-year) in the first ten months of the year. One of the most dynamic aspects of this industry is civil engineering, reflecting the large projects underway to create public infrastructure, notably motorways.

Employment and unemployment

Employment growth rose to record rates in 1999…

Employment growth rose to 5.0 per cent in 1999, the highest rate since 1985 (Figure 3). Most of the new jobs were in the business services and financial sectors. Cross-border workers again accounted for most of the growth in domestic employment. Growth in national employment also strengthened, to 2.5 per cent, mainly reflecting higher employment of immigrants. Registered unemployment continued to edge down, declining by 0.2 percentage point to 2.9 per cent (2.4 per cent on the standardised measure) in 1999. The wider national measure of unemployment, which adds persons in active labour market programmes (special work schemes or training) to registered unemployed, fell by the same amount, to 3.8 per cent in 1999.

… and has been even higher this year

The further strengthening in the economy has pushed the annual rate of employment growth up to 5.5 per cent during the first eleven months of 2000

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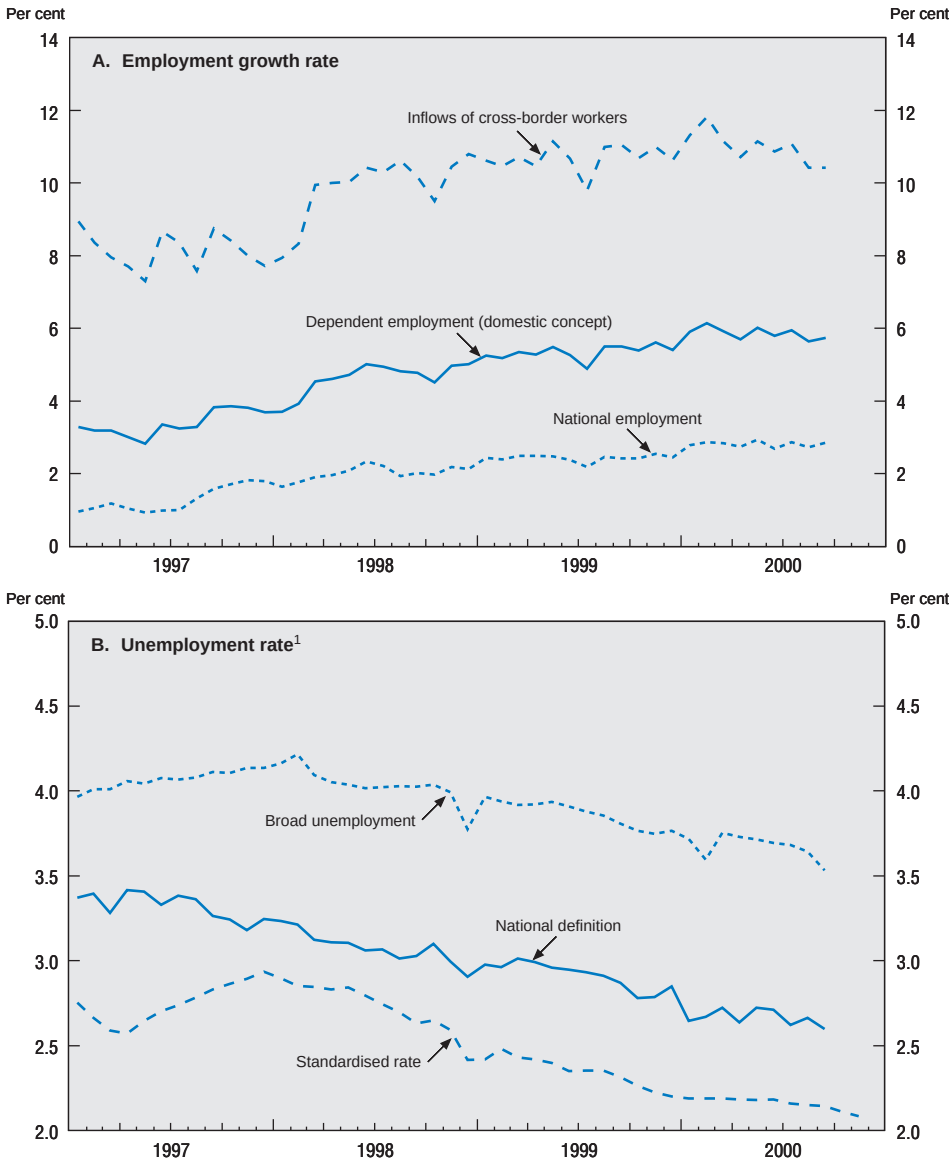
Figure 3. **Employment and unemployment**



1. Seasonally adjusted. The national definition measures job seekers registered with the national employment office. The broad concept adds to registrations persons on work schemes or enrolled in training sponsored by the employment office.

Source: STATEC and OECD, *Main Economic Indicators*.

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(year-on-year). As in 1999, the business services and financial sectors have been the richest in job creation, with annual rates of employment growth of 12 per cent and 8 per cent, respectively, during the first nine months of 2000. Even the manufacturing sector has increased employment in 2000, in contrast to the stagnation of recent years. As usual, cross-border workers have filled most of the additional jobs created, accounting for 60 per cent of the new jobs. Immigrants have also continued to help meet the demand for labour in Luxembourg, contributing to relatively high growth in national employment (2.7 per cent) over this period. As in past periods of exceptionally strong demand for labour, temporary employment has boomed, with hours worked increasing by 16 per cent (year-on-year) in the first nine months of the year. Registered unemployment has fallen slightly over the past year to an average of 2.6 per cent over the three months to November 2000 (compared with 2.8 per cent over the same period in 1999). Unemployment in the wider sense has fallen by the same amont, to 3.6 per cent in the three months to November 2000. The strong labour market has contributed to a reduction in the duration of short-term unemployment. By contrast, there has been a small increase in long-term unemployment, indicating that there are some hard core unemployed whom it is difficult to get back into jobs.

Inflation and wage rates

Rising energy prices have pushed up inflation…

Inflation has increased sharply in the past year, reaching 3.5 per cent (year-on-year) in the fourth quarter of 2000, compared with 2.0 per cent one year earlier (Figure 4). This mainly reflects energy prices, which increased by 32 per cent over this period. Nevertheless, underlying inflation[4] has also increased markedly, from 1.3 per cent in the fourth quarter of 1999 to 2.0 per cent in the third quarter of 2000 and has continued at this rate since then. As a high proportion of goods and services consumed in Luxembourg is imported, most of this increase reflects trends in inflation in other countries and the depreciation of the euro. However, high wage increases since late 1999 have probably contributed to a rise in domestically-sourced consumer price inflation.

… boosting index-linked wage increases…

Wage increases have risen progressively since the first half of 1999. While the available data show that they rose to 4.0 per cent (year-on-year) in the first half of 2000 from 3.0 per cent for 1999 as a whole, this understates the extent of the acceleration as the figure for the first half of 2000 does not take into account the public sector wage agreement of July 2000, which was backdated to January 2000[5] (Table 2). The major factor underpinning this acceleration is wage indexation. An

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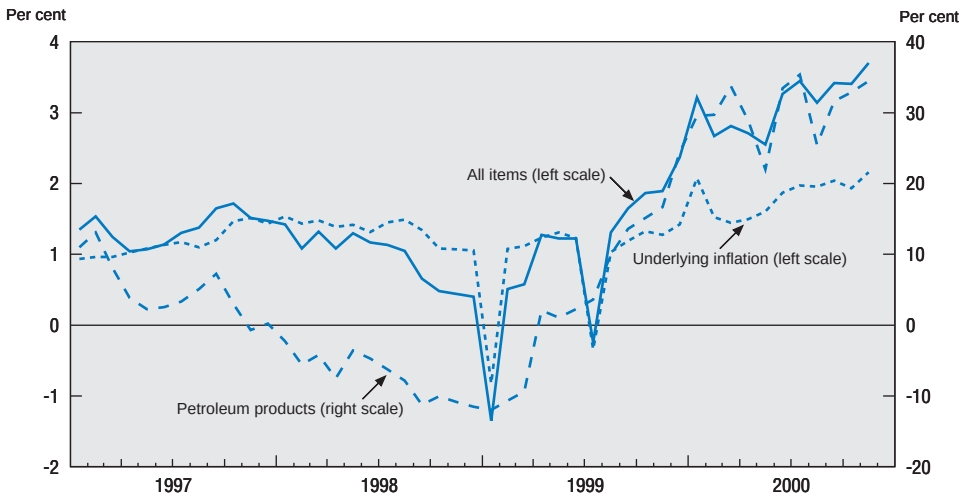
Figure 4. **Inflation**
Per cent change over previous year



Source: STATEC.

indexation threshold was breached for the first time in 2½ years in August 1999, triggering a 2.5 per cent increase in wages and salaries (and social security benefits). Another indexation threshold was breached in June 2000, less than one year since the previous indexed increase, boosting the average increase in wages to 5.7 per cent (year-on-year) in the third quarter of 2000. The next indexation threshold is likely to be breached in the second quarter of 2001.

Table 2. **The contribution of indexation to average wage increases**
Percentage change

	Indexation	Bonuses	Other	Total
1995	1.9	0.5	0.6	3.0
1996	0.8	0.3	0.7	1.8
1997	2.3	0.1	0.4	2.9
1998	0.2	0.7	0.9	1.8
1999	1.0	0.6	1.3	3.0
2000 (6 months)	2.5	0.0	1.6	4.0

Source: Inspection générale de la sécurité sociale (IGSS) calculations by STATEC.

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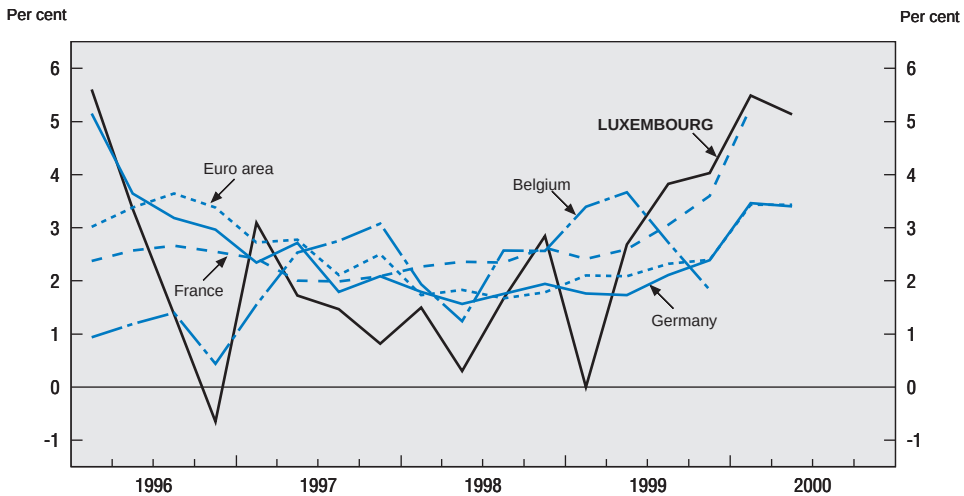
Figure 5. **Standardised hourly wage costs**
Percentage change over previous year



Source: Eurostat.

... and resulting in hourly wage rises amongst the highest in Europe

Wage inflation in Luxembourg was low by comparison with other European countries over 1996-98 but has since accelerated to a relatively high rate (Figure 5). On a standardised basis, increases in hourly wage rates rose to 5.1 per cent in the second quarter of 2000, compared with 3.5 per cent in the euro area. Although this somewhat overstates the situation in Luxembourg because quarterly observations are for the last month in the quarter, whereas they are for all months in most other European countries, the fact remains that Luxembourg now has wage inflation that is high by European standards. Moreover, wage increases are set to rise further owing to the index-linked increase which occurred recently.

Balance of payments

Aircraft purchases reduced the current account surplus in 1999...

The current account surplus fell sharply in 1999, to LUF 52.3 billion (7.1 per cent of GDP) (Table 3). This decline (LUF 11.5 billion) was attributable to one-off purchases of aircraft, which rose by LUF 20.3 billion. These purchases account for the increase in the trade deficit. The terms of trade also deteriorated,

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Table 3. **Current balance of payments**
LUF billion

	1995	1996	1997	1998	1999[1]	1999[1, 2]	2000[1, 2]
Goods and services							
Goods	–46.8	–53.7	–71.4	–71.9	–89.9	–57.6	–62.5
Exports	272.4	262.5	303.2	326.7	320.4	237.2	271.7
Imports	319.3	316.2	374.6	398.6	410.3	294.8	334.1
Services	70.3	77.7	100.2	111.4	131.1	98.2	113.6
Exports	248.4	285.8	349.3	402.5	477.5	347.9	446.4
Imports	178.1	208.2	249.1	291.2	346.4	249.7	332.7
Income, net	65.9	61.8	49.3	37.2	24.3	28.0	24.6
Compensation of employees	–43.8	–49.2	–56.2	–65.2	–77.5	–54.5	–66.0
Investment income[3]	109.7	111.1	105.5	102.4	101.8	82.5	90.6
Current transfers, net	–18.6	–15.5	–18.2	–12.8	–13.2	–13.9	–6.6
Current balance, net	70.7	70.3	59.9	63.8	52.3	54.7	69.1
As a per cent of GDP	13.1	12.5	9.6	9.6	7.1	7.5	11.4[4]

1. Provisional figures.
2. Sum of the first nine months.
3. Including reinvested earnings.
4. Based on projected GDP in 2000.

Source: STATEC and OECD.

by 1.2 per cent,[6] reflecting higher energy prices and falling prices for steel products, but the effect of this on the trade balance was offset by a rise in net export volumes (abstracting from aircraft purchases). The large increase in the surplus on services, especially financial services, attenuated the overall decline in the current account surplus. Financial service exports boomed owing to rapid expansion in funds management services; funds under management grew by 31 per cent in 1999. The transport (notably airfreight) and communications sectors' structural surpluses also grew markedly. By contrast, the net income balance continued to decline, reflecting growing salary payments to cross-border workers.

... but it is set to rise markedly in 2000

The current account surplus has increased markedly this year, to LUF 69.1 billion in the first nine months of the year. In annual terms, the trade deficit is running at around the level it would have been last year in the absence of the unusually large purchases of aircraft. Both exports and imports (abstracting from aircraft purchases) of goods have boomed,[7] reflecting the strengthening in export markets and the high import content of both domestic demand and exports. The surplus on services has again increased sharply, mainly owing to further growth in funds management; funds under management reached € 901 billion in

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the third quarter of 2000, making Luxembourg the second biggest centre in the world (after the United States) for funds management. Service exports also continued to grow rapidly in the transport (especially airfreight) (+15 per cent) and communications (+18 per cent) sectors. The net income surplus has continued to decline, despite a sharp increase in earnings from financial intermediation, owing to rising salary payments to cross-border workers.

Prospects[8]

Monetary conditions and export markets should become progressively less supportive

Monetary conditions in the euro area have been particularly supportive of growth since early 1999, despite an increase in policy interest rates, owing to the substantial depreciation of the euro (Figure 6). It is assumed for the projections presented below that short-term policy interest rates rise further, increasing the 3-month EUROBOR rate from 4.9 per cent in the second half of 2000 to 5.5 per cent one year later, but that they subsequently remain unchanged. This, together with the assumption that the euro exchange rate remains unchanged at its value of 30 October 2000 (US$0.84), implies a tightening in monetary conditions over the projection period. Oil prices are also likely to be a drag on growth, albeit diminishing: they are assumed to remain high, at US$30 per barrel in the second half of 2000 and the first half of 2001, and only to decline to US$27 by the end of the projection period. High oil prices and tightening monetary conditions underpin the assumed slowing in export market growth; economic growth in the EU is assumed to slow from a cyclical peak of 3.3 per cent in 2000 to 2.7 per cent in 2002. On the other hand, discretionary fiscal policy is set to ease. The government has announced personal and corporate income tax cuts amounting to 1.2 per cent of GDP in 2001 and an additional 1.6 per cent of GDP in 2002 (of which personal income tax cuts amount to 0.6 per cent of GDP). It has been estimated that the household tax cuts (LUF 10 billion in 2001 and a further LUF 5.5 billion in 2002) could lead to an increase in real private consumption expenditure of 4.8 per cent spread over four years (STATEC 2000*b*, p. 31). As a result of these tax cuts, the general government surplus is projected to decline from 4½ per cent of GDP in 1999 to 2½ per cent of GDP (on a cash basis) in 2002.

Growth and inflation should slow from a peak in 2000

Real GDP growth is projected to hit a cyclical peak in 2000, at 8 per cent, and to slow to 5¾ per cent by 2002, slightly above the long-term average rate (see Table 1). As noted above, euro depreciation and strong export market growth have contributed to the surge in growth but these stimuli are assumed to wane over the projection period. Employment growth should follow the pattern for economic growth, peaking at around 6 per cent in 2000 and declining to a still

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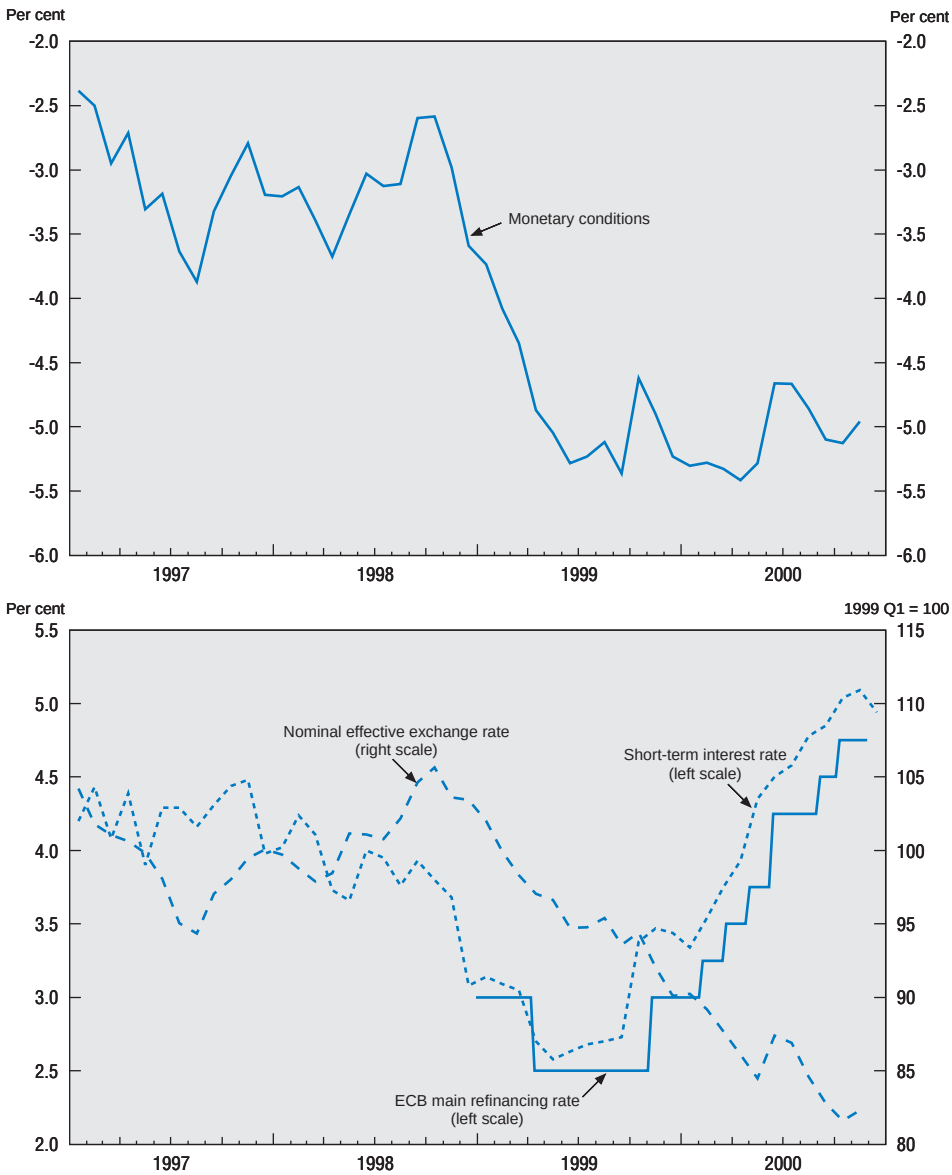
Figure 6. **Monetary conditions**[1]
Per cent



1. The weights underlying the index are 1 for the short-term interest rate and 0.15 for the exchange rate.
Source: European Central Bank (ECB), *Monthly bulletin.*

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robust 4 per cent by the end of the projection period. As usual, cross-border workers should fill most new jobs and, together with net immigration, bear most of the burden of adjustment to changes in employment growth. In these circumstances, the (registered) unemployment rate is projected to edge down only slightly, to about 2¼ per cent in 2002, despite robust employment growth. Inflation is likely to peak at 3 per cent or so in 2000 and to fall back to about 2 per cent in 2002. The main risk to these projections is that the energy-price shock could lead to a wage-price spiral, which could occur more quickly than in other countries owing to indexation arrangements.

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II. Fiscal policy

Overview

Luxembourg's fiscal performance compares very favourably with that of other OECD countries, with a budget in surplus almost uninterruptedly over the last two decades. High growth rates of real GDP, averaging over 5 per cent, supported by a rapidly expanding financial sector, generated increasing government revenues and enabled successive governments to expand the public sector to a correspondingly high level, and to build up a relatively generous system of social security. The strong increases in receipts also allowed for the building up of government assets, which are to be used in the future to pay part of the pensions and social security benefits to the rapidly rising number of employees, including cross border workers.

In view of the ageing population, over the past few years fiscal policy was rightly focused on maintaining surpluses, both by restraints on expenditure and by the promotion of economic growth, which should generate additional revenues. Moreover, although maintaining such surpluses in the case of Luxembourg might look like going beyond the requirements of the Pact for Stability and Growth (to respect safety margins below the deficit ceiling of 3 per cent of GDP), a large margin seems desirable for Luxembourg to take into account the risks related to the economy's high degree of specialisation.

Large surpluses on the budget balance were recorded in 1998 and 1999 and are likely to continue in 2000 and 2001, as economic prospects are good. Developments in 2001 and beyond will also be affected by the recently announced tax reform the government intends to implement in 2001 and 2002, and on further expenditure restraint. Structural improvements, directed toward diversification of the economy, remain important objectives as well, given the heavy dependence of government revenues on the financial sector, which in the medium term might have to face more intensified competition with other financial centres.

In addition, it should be noted that the surplus is becoming increasingly dependent on surpluses generated in the social security sector, notably the pension fund. Were this trend to be reversed, be it as a result of a shift to a lower

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growth path of GDP in the medium term, or owing to the population ageing in the future, other components in the overall budget balance would need to be sufficiently flexible to prevent the overall surplus from falling too fast or too far.

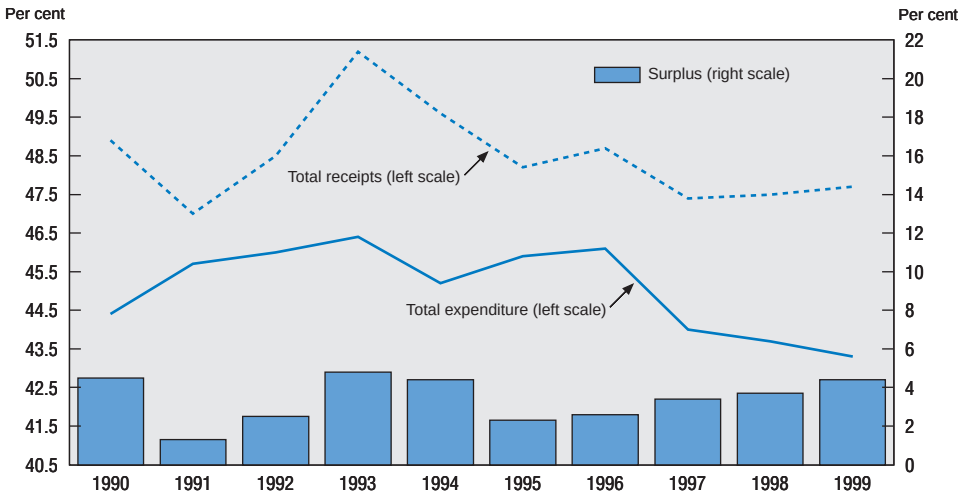
In view of the potential for the budgetary burden of pensions to grow in coming decades, the report that the authorities have commissioned on pension financing, including the scope for greater reliance on private pension funds, is to be welcomed. While increased reliance on pension funds for the provision of retirement income would entail considerable transition costs, such a reform would enhance the long- term sustainability of fiscal policy in Luxembourg.

Recent developments

In 1999, the surplus on the general government account continued to trend upward and came out at 4.4 per cent of GDP, which was some 2 percentage points higher than in 1995, at the start of the improvement (Figure 7).[9] Due to an unexpected cyclical improvement in 1999, the outcome was also more favourable than projected in the Stability Programme of early 1999.

The increase in the surplus in 1999, by 0.7 percentage point, was the result of both a rise in the ratio of revenues to GDP (to 47.7 per cent) and a fall in

Figure 7. **General government finances**
As a percentage of GDP



Source: STATEC.

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the expenditure ratio (to 43.3 per cent). A strong rise in wage earnings, due to the cyclical upswing, resulted in correspondingly higher revenues from direct taxes. The rise in social security contributions was also the result of an increase in contribution rates to finance a new long-term care insurance (*l'assurance dépendance*) introduced in 1999. Boosted by a higher volume of private consumption and higher energy prices, indirect taxes increased by 16 per cent. This revenue category now accounts for about one-third of total government revenues and its share in GDP is one of the highest in the OECD. By contrast, corporate taxes, which in the past ranked very high as well, continued to trend downward, as a result of tax cuts aimed at promoting economic activity and enhancing the attractiveness of Luxembourg as a location for enterprises.

Government expenditure increased by 8½ per cent. The government succeeded in keeping the growth of nominal compensation of government employees subdued (at 5.6 per cent), both by limiting the increase in wages, which rose primarily as a result of the automatic indexation in August 1999, and by restraints on employment. Social security benefits and government investment also grew less rapidly than GDP. Some expenditure categories, however, showed very strong rises. The introduction of *l'assurance dépendance* accounted for additional expenditure of LUF 4.6 billion (0.6 per cent of GDP). Spending on health care, family allowances and pensions was also higher.

A law was passed by Parliament in 1999 with the aim of improving efficiency and transparency in the government sector itself. Measures to modernise Luxembourg's budgetary and accounting system included improvements in budgetary accounting rules, the introduction of commitment-based accounts and the creation of the post of Financial Controller. In addition, the possibility was introduced to grant certain government departments greater financial autonomy. Finally, measures were taken to speed up tax payment procedures.

The budget in 2000

In the Coalition Agreement of 7 August 1999, three main principles were formulated for prudent and sound management of public finance.

- The general government balance should remain in surplus.
- The balance of central government should remain in equilibrium.
- Current expenditure of central government should increase less rapidly than total government expenditure and nominal GDP; this is one of the key objectives of the new government.

In 2000, all three objectives are likely to be achieved. In the February 2000 Stability Programme three scenarios were presented, with average growth rates of GDP in Europe ranging from 2.5 to 3.5 per cent and with corresponding growth rates in Luxembourg varying from 4.3 to 6.0 per cent. When the programme was

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prepared, it was already clear that economic activity in Luxembourg had not yet entered a phase of deceleration. Nevertheless, for the sake of prudence, the low growth scenario, implying GDP growth in 2000 of some ½ percentage point less than in 1999, was chosen as a basis for macroeconomic and fiscal projections. In this scenario, the budget surplus was projected to rise by 0.2 percentage point. Recent figures on employment growth, however, point to a growth rate of GDP, which is even slightly higher than in 1999. Despite this cyclical effect, the budget surplus is set to fall slightly to around 4 per cent of GDP, as the positive cyclical effect will be offset by a setback in capital receipts (Table 4). All three sub-sectors of general government are likely to contribute to the surplus in 2000, including the central government, which in recent years already showed better results than the balance formulated as a minimum target for 2000. However, not too much weight should be attached to the borderline between individual sub-sectors and to the size of their surpluses, given the large transfers between them. With respect to the third "target" mentioned above, limiting spending by central government, current expenditure should fall due to stabilisation of the number of civil servants, thus creating room for an increase in government investment in infrastructure, schools and hospitals, as well as for further tax reductions. Consolidated expenditure of total government should not grow rapidly either. Transfers of pensions are projected to fall, as a share of GDP, but spending on health services may continue to show some upward pressure. In early 2000 this led the government to raise contribution rates.

All in all, the outlook for public finance in 2000 is bright. However, it should be noted that the surplus probably contains a significant positive cyclical component, owing to the very high growth rates in recent years (about 7 per cent on average in the period 1997-2000).

Table 4. **General government budget and debt**

	1996	1997	1998	1999[1]	2000[2]
General government net lending (LUF billion)	14.6	21.4	24.9	32.2	32.3
As a percentage of GDP					
General government	2.6	3.4	3.7	4.4	4.0
Central government	0.9	2.0	2.0	2.7	..
Local government	0.7	0.5	0.3	0.1	..
Social security	1.0	0.9	1.5	1.5	..
General government gros debt (LUF billion)[3]	34.7	37.5	42.8	43.6	45.0
As a percentage of GDP	6.2	6.0	6.4	6.0	5.5

1. Provisional.
2. Estimates.
3. Maastricht definition.

Source: STATEC and Banque centrale du Luxembourg.

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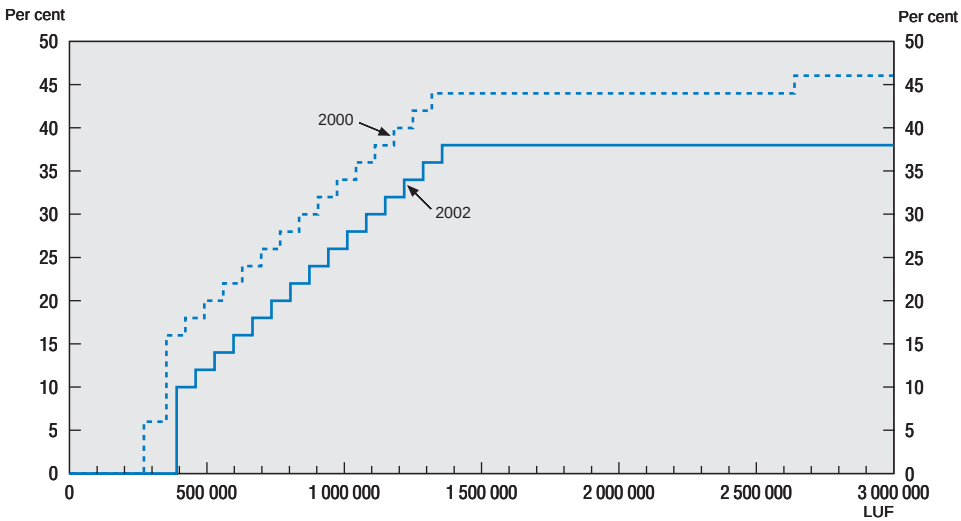
The budget in 2001 and beyond

A first impression of budgetary developments in 2001 is provided for by the budget memorandum presented by the central government on 22 August 2000. The most important change announced, to be approved by Parliament before the end of 2000, is a tax reform in 2001 and 2002, which will reduce the budget surplus by some 1 percentage point of GDP in 2001 and by another 2 percentage points in 2002 from what it would otherwise have been. However, with growth projected to remain high, the budget surplus (on a cash basis) should only fall to around 2½ per cent of GDP in 2002.

The planned reduction in the tax burden for both households and enterprises is aimed at further increasing the competitive position of Luxembourg and creating new incentives for working-age people to participate in the labour force. The first phase of the tax reform, benefiting households, is planned to take place in 2001. It includes a substantial increase in the exemption threshold in personal income taxes (Figure 8) and a reduction in tax rates by 2 percentage points. In addition, the maximum (marginal) rate will be reduced from 46 to 42 per cent. In 2002, as a second step, most tax rates will be reduced by another 4 percentage points, so that the highest (marginal) rate will fall to 38 per cent.

Figure 8. **Marginal tax rates on household income before and after the tax reform**[1]



1. Single (class 1).
Source: Finance Ministry.

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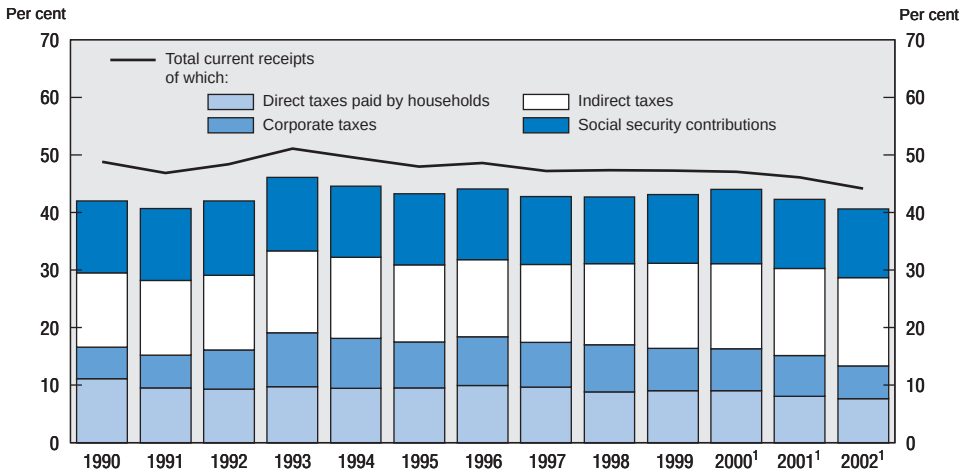
The tax reform may lead to a gain in disposable income of households of 2.4 per cent in 2001 and 3.6 per cent in 2002 relative to the baseline scenario.[10] The total effect on household consumption, cumulated over four years, is estimated to be a little less than 3 per cent (80 per cent, the marginal propensity to consume, of the cumulated rise in disposable income). Given the share of private consumption in GDP, which in Luxembourg is about one half, the total rise in expenditure by households would correspond to an average effect on GDP of about 0.4 per cent a year over the period 2001-04.

Not surprisingly, the tax relief for households will lead to a significant loss of government revenues: 1.2 per cent of GDP in 2001 and another 0.6 per cent in 2002. Moreover, in 2002 the corporate tax rate (including the municipal business tax, the *impôt commercial communal*) will be cut as well, from the current 37.45 per cent to 30 per cent. This may lead to a further fall in revenues (on an accruals basis) by about 1 per cent of GDP (Figure 9). Nevertheless, total direct taxes (on a cash basis) may only decline by about 1¼ per cent of GDP between 2000 and 2002 owing to high growth. On balance, the general government surplus (on a cash basis) is set to fall to 2½ per cent of GDP in 2002.

The tax relief implies a significant stimulus to private expenditure at a time when the economy is already growing strongly. Therefore, some upward

Figure 9. **Composition of current receipts**
As a percentage of GDP



1. Estimates.
Source: STATEC and OECD.

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pressure on prices is likely, although many consumption goods are imported from abroad or bought in border areas in Luxembourg itself, where competition between domestic and foreign retailers helps to keep inflation subdued. Increases in purchasing power might also be directed towards the housing market, leading to higher prices of houses and land for building.

Total government expenditure in 2001 is projected to grow about as rapidly as GDP, *i.e.* at a rate far above trend growth. The budget for the central government contains significant rises in spending on infrastructure projects, notably road construction, on education, and on research and development. Social benefits will rise even more rapidly than GDP. However, these increases as a share of GDP will be offset by a decline in compensation of government employees as a percentage of GDP, despite a generous increase in wages of 40 000 public-sector workers (a rise of 3.5 per cent plus indexation for inflation, increases in travel allowances and one more holiday day a year). Civil servants' pensions are also set to increase only moderately, due to the reform introduced by the previous government.

Medium and long-term prospects

The prospects for public finances in the medium term in Luxembourg are good. Owing to the continuous stream of budget surpluses so far, net financial assets of the government sector have steadily increased to well over 30 per cent of GDP and in the near future gross consolidated debt will remain under its peak of 6.4 per cent of GDP, recorded in 1998.

In the long run, the generous pay-as-you-go pension scheme[11] could result in serious budgetary pressures. Assuming that over the next five decades employment and labour productivity were to continue to grow respectively at an annual rate of 3 per cent and 2 per cent, the averages of the past 15 years, the IMF has calculated that pension expenditures could be held to their present share of GDP (9 per cent; Figure 10, Panel A).[12] This would imply sustained massive inflows of cross border workers, increasing their number to six times that of resident workers by 2050. A less favourable scenario, also analysed by the IMF, is based on a stabilisation of the proportion of cross-border workers in total employment (40 per cent) from 2005 onwards. In this scenario, there is no employment growth in 2005-50, reflecting the demographic characteristics of the resident population. Pension expenditure would then rise to around 22 per cent of GDP in 2030. Such an increase is very large in relation to the government's existing net financial assets.

Assuming the less favourable scenario, based on roughly the same assumptions as those of the IMF, but also taking into account the current stock of net financial assets, it can be shown that if no measures are taken, both the budget and the net financial asset position of general government would deteriorate dramatically. By 2020, the budget would show a deficit of about 13 per cent of

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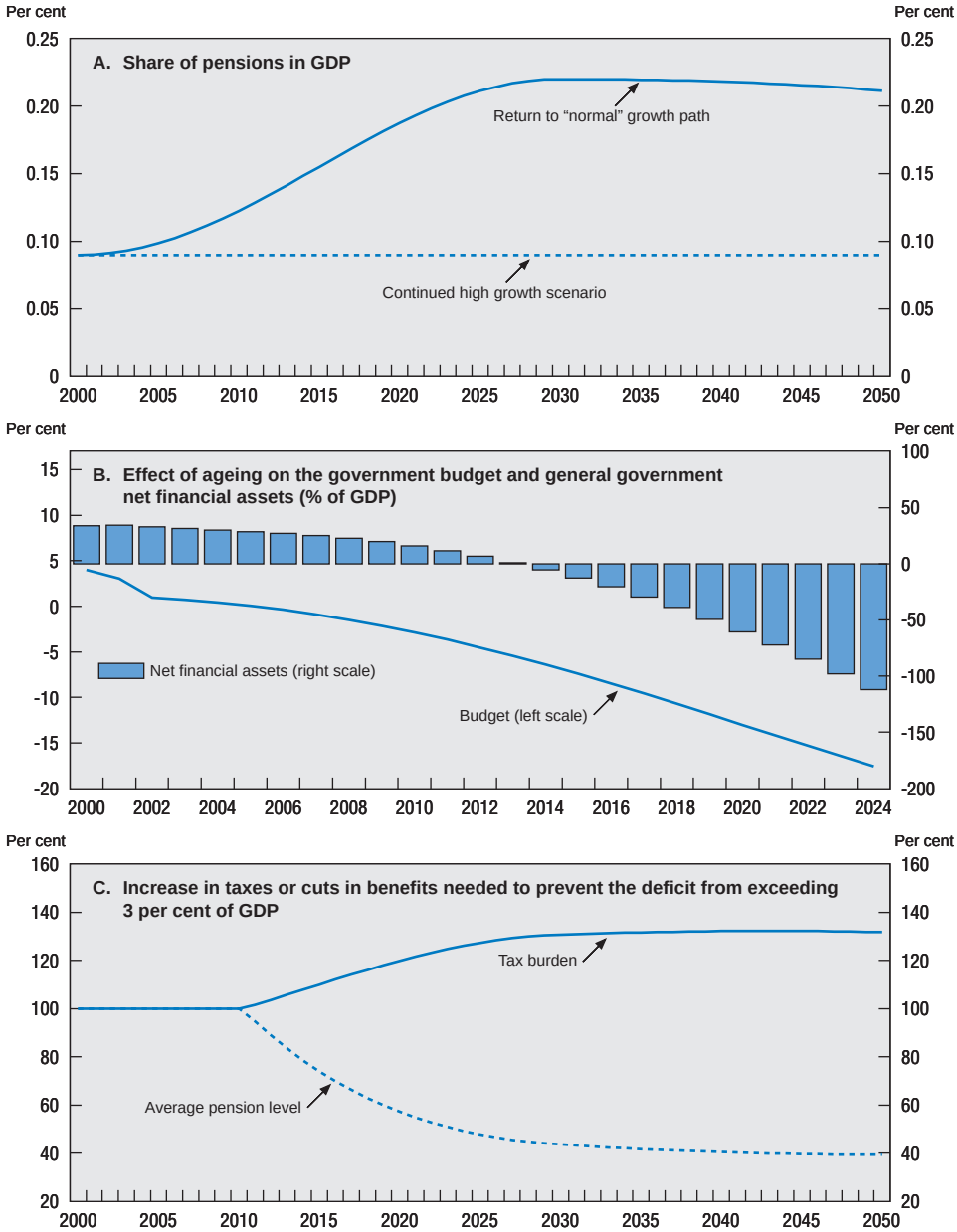
Figure 10. **Ageing and public finances**



Source: STATEC and OECD.

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GDP, far above the ceiling of 3 per cent to be respected according to the Stability and Growth Pact (Figure 10, Panel B). Moreover, net financial assets of the government would have fallen to minus 60 per cent of GDP, with net debt continuing to rise rapidly in subsequent years. This highlights the fact that the apparently substantial net financial assets at present are small in relation to the potential future budgetary pressures from pensions. These assets represent only 4 per cent of the present value of cumulated pension liabilities in the period 2001-50.[13] Alternatively, the authorities could try to prevent the deficit from exceeding 3 per cent of GDP. This would require either increases in government revenues, by about one-third (some 15 per cent of GDP), or reductions in the average level of pension benefits by as much as 60 per cent (Figure 10, Panel C) or a combination of the two.

Although changes aimed at limiting the budget deficit to 3 per cent of GDP might be phased in gradually, they would conflict with major existing policy objectives such as a reduction of the share of the public sector in GDP, enhancing the attractiveness of Luxembourg through tax cuts, making public finances less vulnerable to external shocks and securing stable retirement incomes. In view of these concerns, the authorities have commissioned a report on pension financing which will address the issue of whether there should be greater reliance on private pension funds for the provision of retirement income. While moving in this direction would entail transitional costs because of the need both to finance current pension benefits and to contribute to funds for future benefits, it would enhance the long-term sustainability of fiscal policy in Luxembourg. Such a reform would also increase the return on pension contributions if the return on fund investments were higher than the growth of the total wage bill. Increasing the effective retirement age might also help in coping with the ageing problem (transition ages to retirement in Luxembourg rank among the lowest in the OECD: 58 years for men, and 55 years for women).

Funded pension schemes should also be considered for civil servants, after the steps taken in recent years to reduce public sector pensions (from 83.3 per cent of final income to 72 per cent) and after the significant increase in contribution rates since the early 1990s. Furthermore, management of the public sector's financial assets, as well as the surpluses in private sector pension schemes (for which a study has already been commissioned), should be improved. Investments in a broader range of investment products might contribute to higher rates of return and thus contribute to a helpful – albeit not sufficient – buffer to deal with the increasing pension burden in the future.

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III. Progress in structural reform

Luxembourg's economic performance during the past two decades has been remarkable. As in surrounding regions, Luxembourg was severely affected by the sustained decline in the steel industry that began in the 1970s. However, in contrast to its neighbours, Luxembourg was able to switch rapidly to high value added services, notably financial services. An important factor that enabled Luxembourg to diversify in this way was its regulatory, taxation and other advantages which its neighbours do not share. As these industries expanded, growth was reinforced by the external benefits of clustering. This transformation of the Luxembourg economy underlies the extraordinarily high economic growth (5½ per cent per year) and low unemployment recorded since the mid-1980s.

These favourable developments, however, mask some structural weaknesses in the Luxembourg economy. The acceleration in growth since the mid-1980s has been achieved through an increase in growth in productive inputs: multi-factor productivity growth has actually slowed. While rapid growth in employment has sheltered residents from the large increases in unemployment that occurred in other European countries with similar labour market institutions, it has not resulted in a significant increase in the employment ratio mainly owing to inappropriate barriers and disincentives to working. Moreover, the high degree of specialisation of the Luxembourg economy, which is an inevitable feature of a small regional economy, entails the risk that any shift in demand away from the goods and services provided by the leading sectors would have serious consequences for the economy as a whole. Given that the labour market continues to have many employment unfriendly institutions, including high replacement rates for unemployment benefits and, in certain circumstances, for social assistance, which continues for an indefinite duration, such a shock could be expected to increase unemployment substantially and durably. At the same time, a lasting reduction in the employment growth rate would have serious consequences for the sustainability of the pay-as-you-go (PAYG) pension system, as discussed in Chapter II. These considerations suggest that structural reforms continue to be required both to enhance economic efficiency and to prepare for possible adverse shocks in the future.

After briefly examining trends in productivity growth, this chapter provides an overview of key strengths and weaknesses in labour market outcomes and insti-

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tutions and suggests reforms that could contribute to better labour market outcomes. In the following section, some key issues in product markets are discussed, including what Luxembourg has done to facilitate the development of electronic commerce and what still needs to be done. This is followed by a discussion of some of the challenges that Luxembourg faces in establishing framework conditions propitious for the diffusion of information and communications technology (ICT). The chapter finishes with a discussion of regulatory developments that concern the financial sector, including international co-operation on tax matters.

Productivity growth

Multi-factor productivity growth (MFP), which measures growth in output not accounted for by growth in factor inputs, has progressively declined in the market sector excluding banks[14] from a peak annual rate of 2.9 per cent in the 1985-87 productivity cycle[15] to 0.9 per cent in the 1995-99 productivity cycle[16] (Table 5 and Figure 11). MFP growth rates in the 1990s have been markedly lower than the long run average (1.9 per cent since 1975). Growth in labour productivity weakened by much less than MFP growth over 1987-95 because the contribution from capital deepening increased, but has since fallen sharply, reflecting the decline in both MFP growth and the contribution from capital deepening. The increase in economic growth since the mid-1980s has been achieved through an increase in growth in productive inputs, not through an increase in the rate at

Table 5. **Productivity growth**[1]
Average annual growth rate

Productivity cycles	Labour productivity	Capital productivity	Multi-factor productivity	Capital labour ratio	Contribution of capital deepening to labour productivity	Value added, volume
1975-80	2.3	2.4	2.4	–0.1	–0.1	2.5
1980-85	1.9	2.3	2.1	–0.4	–0.2	2.1
1985-87	2.3	3.5	2.9	–1.2	–0.6	4.7
1987-91	2.1	2.7	2.4	–0.6	–0.3	6.4
1991-95	2.1	0.5	1.3	1.6	0.9	3.8
1995-99	0.3	1.4	0.9	–1.1	–0.6	4.6
Longer term						
1975-85	2.1	2.4	2.2	–0.3	–0.1	2.3
1985-99	1.6	1.8	1.7	–0.2	–0.1	4.9
1975-99	1.8	2.0	1.9	–0.2	–0.1	3.8

1. Market sector (including imputed rentals on owner-occupied housing) excluding the banking sector. Labour input is measured in hours.

Source: STATEC and OECD.

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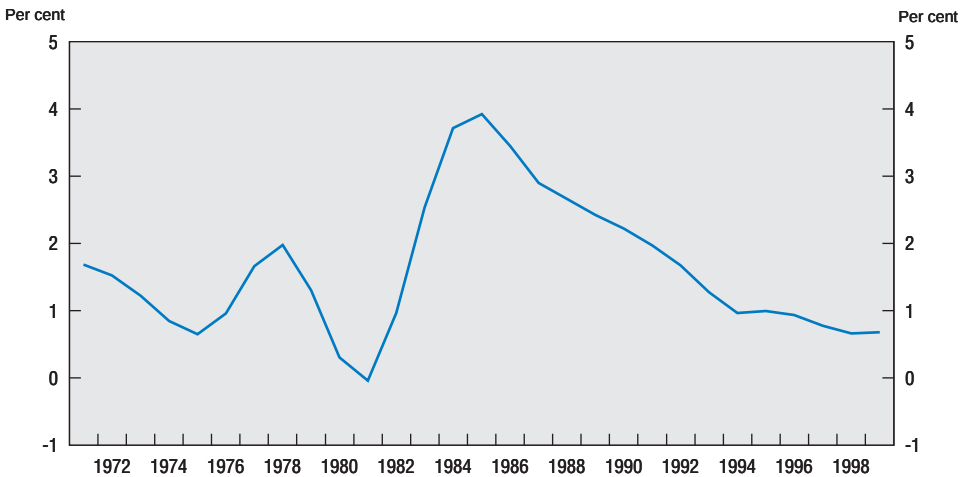
Figure 11. **Trend multi-factor productivity growth**[1]
Growth over previous period



1. Calculated using Henderson 11-period moving average. See note in Table 5.
Source: STATEC and OECD.

which economic efficiency improves. These considerations point to the need for structural reform, including policies to develop the information society, to raise economic efficiency, despite Luxembourg's high economic growth rate and already high level of national income per capita (Box 1).

Labour markets

High employment growth and low unemployment

In some respects, Luxembourg boasts very favourable labour market outcomes. Employment growth has averaged 2.4 per cent per year since 1980, the highest rate in the OECD and far higher than the EU average of 0.4 per cent per year (Figure 13). And the unemployment rate (standardised measure) has remained low by international comparison (Figures 14 and 15). Employment has been buoyed by sustained high rates of economic growth, notably in the financial services and the communications and transport sectors. High growth in these sectors has more than offset the negative effects on employment demand of restructuring in the steel industry and, as in other OECD countries, a productivity slowdown, a rise in real interest rates and an adverse shift in demand for labour.[17] Thus, Luxembourg has not experienced the sharp slowdown in labour demand

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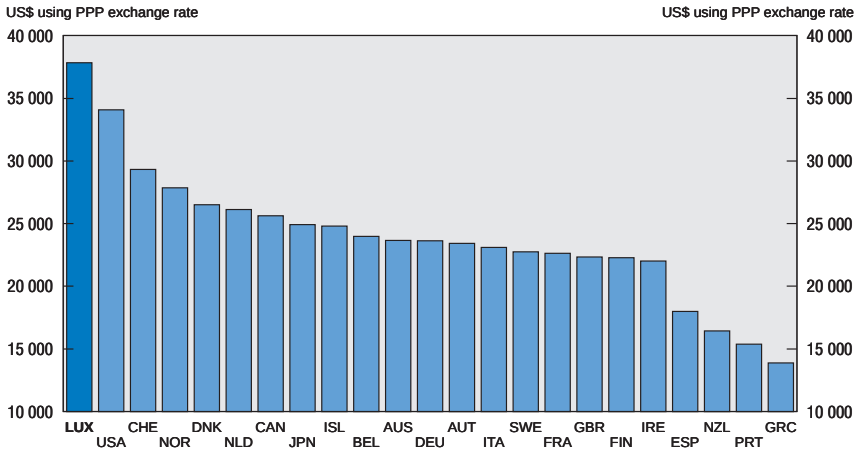
Box 1. Measuring living standards

Commentators frequently use GDP per capita (at purchasing power parity (PPP) exchange rates) to compare living standards across countries. Such an approach is not appropriate for countries that have substantially different net factor payments to the rest of the world. This consideration is important for Luxembourg, as there are large salary payments to non-residents (*i.e.* cross-border workers) as well as substantial net receipts of capital income (see Chapter I). For this reason, it is preferable to compare Gross National Income (GNI) per capita (at PPPs). On this basis, Luxembourg is the richest country in the OECD (Figure 12). Nevertheless, this measure, like GDP per capita, is still subject to many weaknesses. In particular, a given level of GNI (or GDP) per capita will reflect different combinations of inputs and efficiency in different countries. A country that uses fewer inputs more efficiently to achieve a given level of GNI (or GDP) per capita will have a higher living standard than one that uses more inputs but less efficiently. Moreover, as individuals, not countries, enjoy living standards, a meaningful assessment cannot be made without taking into account the distribution of income. As making an in-depth comparison of living standards is beyond the scope of this chapter, the data on GNI per capita (at PPPs) are simply presented as an indicator of living standards along with suitable health warnings.

Figure 12. **Gross national income per capita, 1999**[1]



1. 1998 for Australia, New Zealand, Portugal and Switzerland.
Source: OECD, *Labour Force Statistics* and *National Accounts.*

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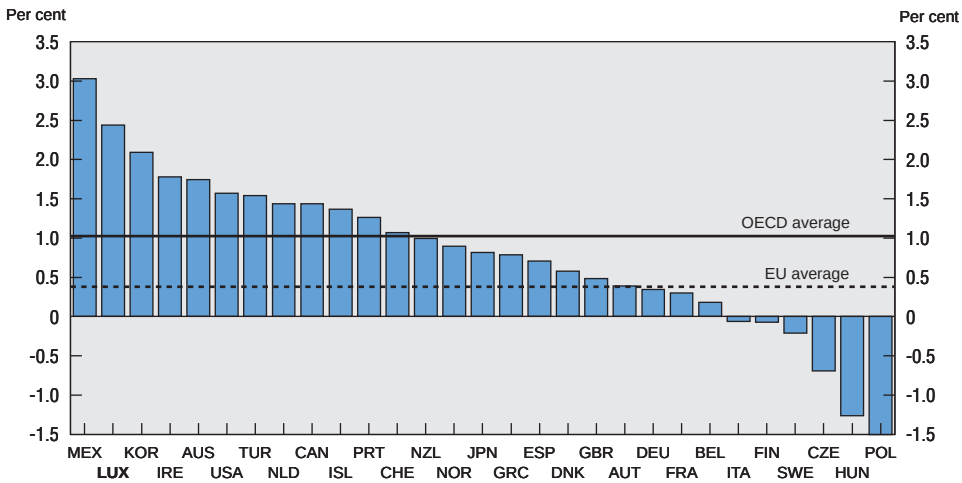
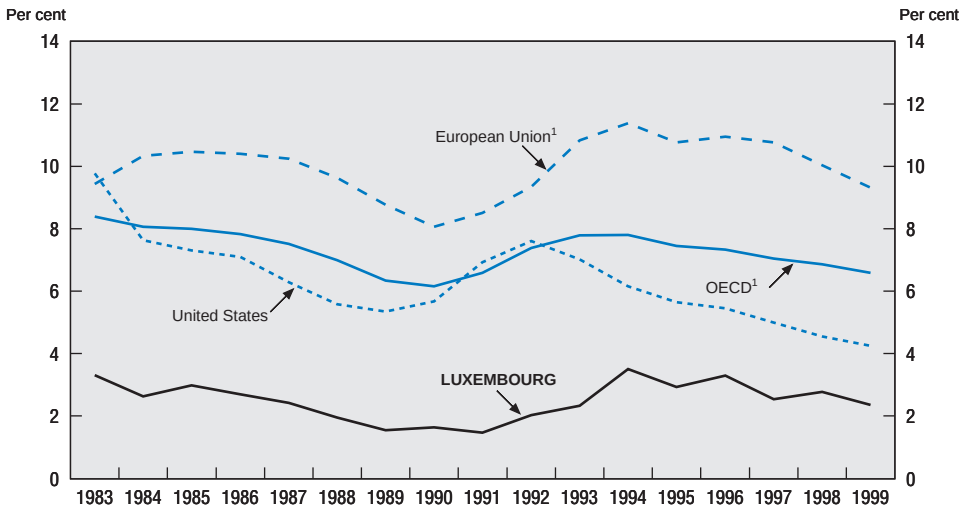
Figure 13. **Employment growth rate**[1]
1980-1999



1. Domestic concept.
Source: OECD.

Figure 14. **Evolution of unemployment**



1. Simple average.
Source: OECD, *Labour Force Statistics*, Part III.

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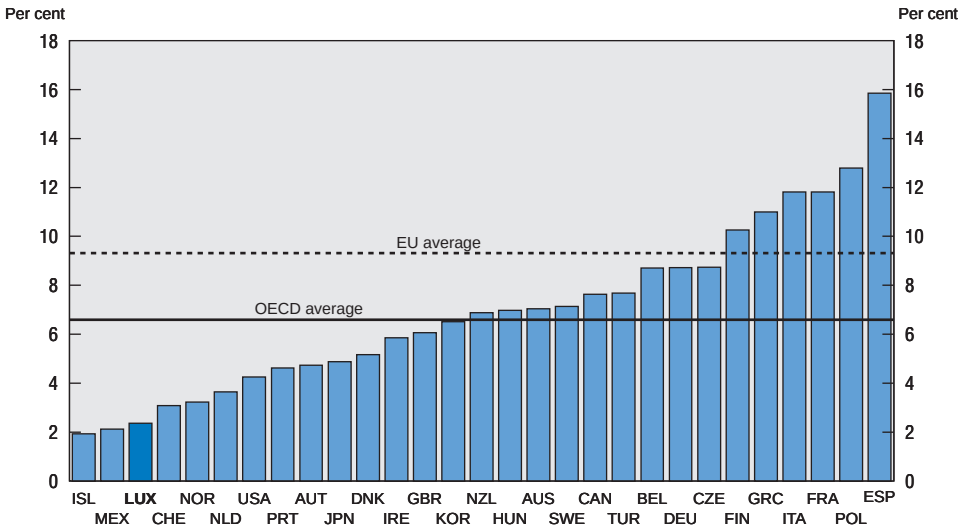
Figure 15. **Unemployment rate, 1999**



Source: OECD, *Labour Force Statistics*, Part III.

that, combined with employment-unfriendly labour market institutions, caused the rise in European unemployment since the 1970s (Blanchard and Wolfers, 1999). An extension of Blanchard and Wolfers' model to Luxembourg confirms that exceptionally buoyant economic conditions explain the paradox of low unemployment despite rigid labour market institutions that in other countries were associated with lasting increases in unemployment (IMF 2000*a*, "Luxembourg's Labour Market Paradox", pp. 27-34). Concomitantly, were a substantial and lasting slowdown in economic growth to occur, unemployment could be expected to rise considerably, as has already occurred in other European countries with similar labour market institutions.

Employment growth in Luxembourg over 1980-99 was far in excess of labour force growth (1.0 per cent per year). This contrasts with the situation in the EU as a whole, where employment growth was less than labour force growth (0.6 per cent). The high rate of employment growth in Luxembourg relative to labour force growth has only been possible owing to rapid growth in employment of cross-border workers (Figure 16). Such employment has grown at an average annual rate of 13 per cent since 1980, increasing the proportion of cross-border workers in domestic employment from 4 per cent to 28 per cent in 1999. Buoyant labour market conditions have also attracted immigrants, boosting labour force

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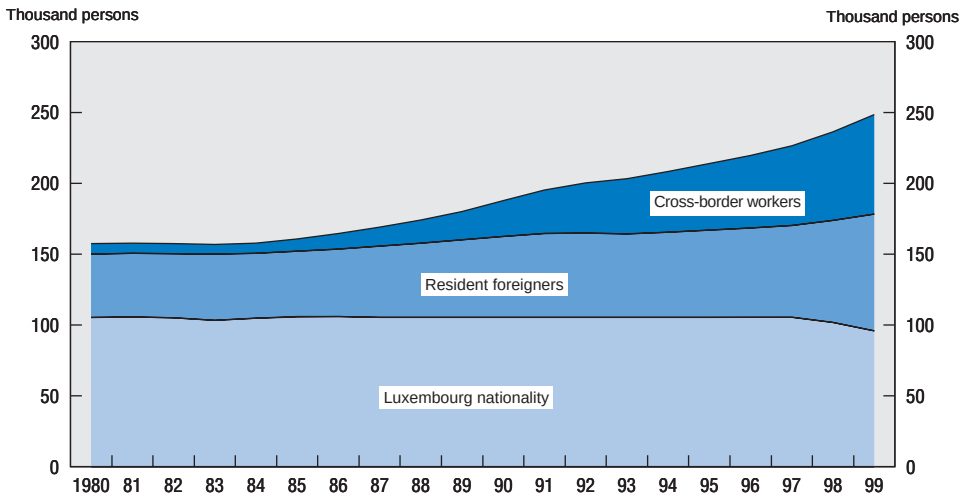
Figure 16. **Decomposition of domestic employment**
1980-1999



Source: STATEC.

growth. All in all, cross-border workers and resident foreigners now account for around 62 per cent of total employment in Luxembourg.

Cross-border workers and resident foreigners also contribute to favourable labour market outcomes by acting as a shock absorber: variations in total domestic employment are disproportionately reflected in variations in employment of cross-border workers and of resident foreigners.[18] Hence, cross-border workers and resident foreigners make labour supply more elastic. This reduces wage pressures when labour demand rises and reduces flows into unemployment when labour demand falls, attenuating the burden of adjustment in the national labour market. This makes Luxembourg less dependent on flexible labour-market institutions than other countries to ensure that cyclical fluctuations in labour demand do not result in economic expansion being cut short or structural unemployment rising.

Low employment ratios, notably for older workers and women

The main weakness in Luxembourg's labour-market performance is that the proportion of the population in employment is not very high, reflecting inappropriate incentives and other barriers to employment (see below) (Figure 17).

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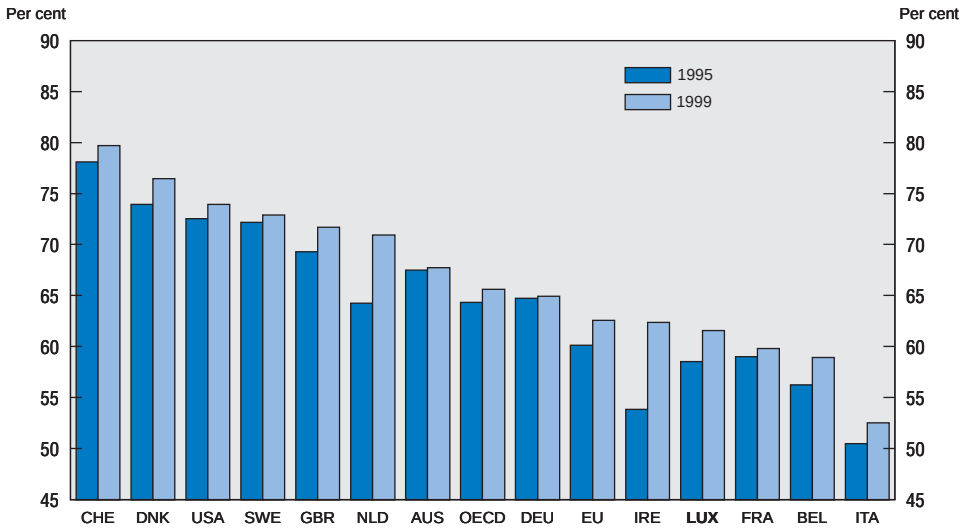
Figure 17. **Employment-population ratio**



Source: OECD, *Labour Force Statistics*, Part III.

While the employment ratio has risen by more than in most other countries in recent years, at 61.6 per cent in 1999 it was nevertheless still slightly below the EU average of 62.6 per cent, which in turn was low by international comparison. By contrast, some one third of OECD countries had employment ratios of 70 per cent or more in 1999. Luxembourg, like most other EU countries, still has a long way to go to meet the objective agreed at the Luxembourg summit of raising the employment ratio to 70 per cent by 2010.

Employment ratios are low by international comparison for all demographic groups except prime-age males (Figure 18), as in other countries with relatively low employment ratios. This is especially so for older workers, for whom the employment ratio was only 26 per cent in 1999, despite having increased in recent years by more than in most other countries; the average ratios in the EU and OECD were 38 per cent and 49 per cent, respectively (Figure 19). The female employment ratio (49 per cent in 1999) is also low compared with the EU and OECD averages of 53 per cent and 55 per cent, respectively (see Figure 18), giving rise to a relatively large gender gap (26 percentage points) in the employment ratio (Figure 20). This nevertheless represents a marked improvement over recent years, the female employment ratio having risen by 6 percentage points since 1995 while the male employment ratio has been stable. Even so, Luxembourg still has a long way to go to achieve the target agreed at the Luxembourg summit of a 60 per cent female employment ratio by 2010.

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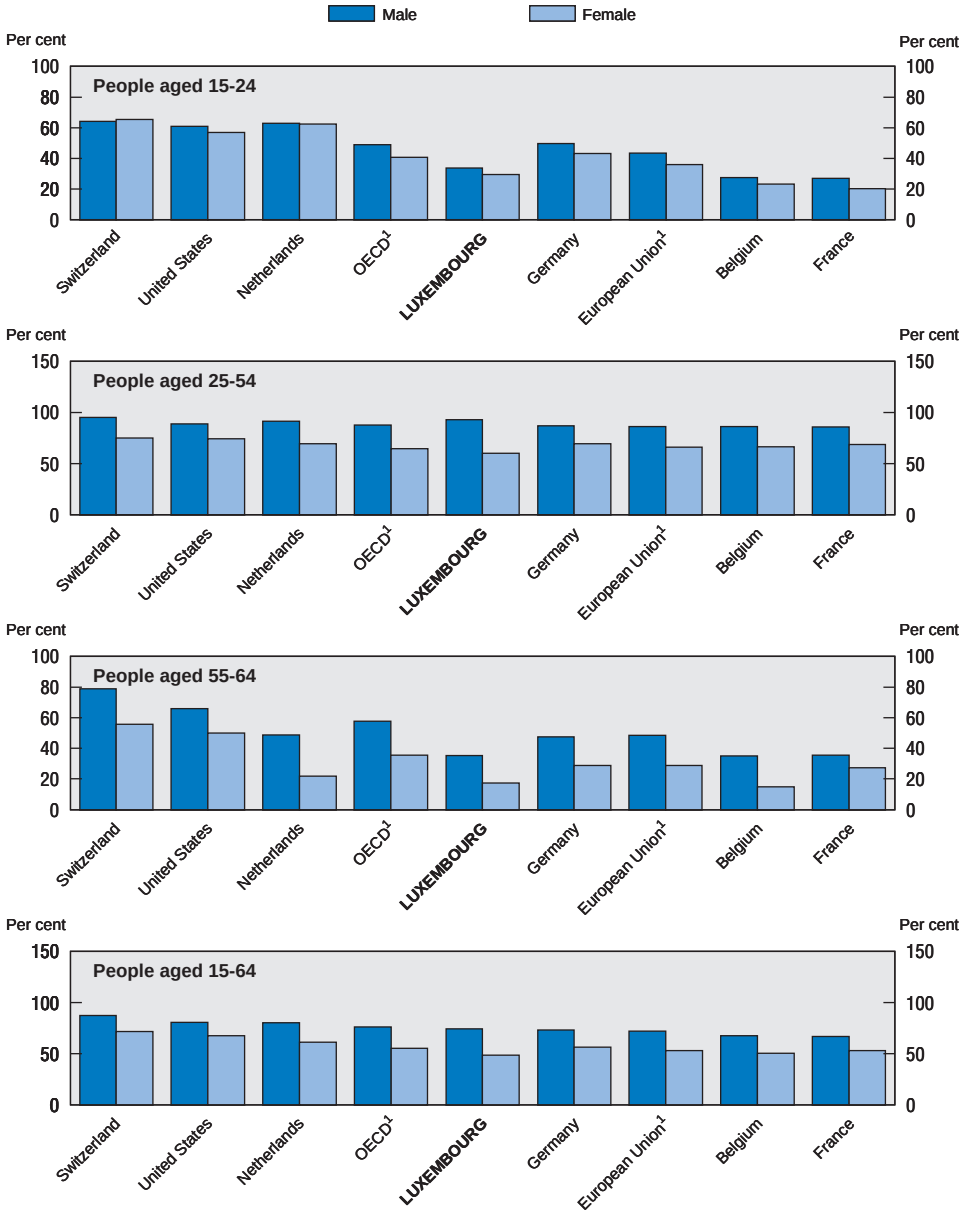
Figure 18. **Employment-population ratio, by gender and age group, 1999**



1. Simple average.
Source: OECD, *Labour Force Statistics*, Part III.

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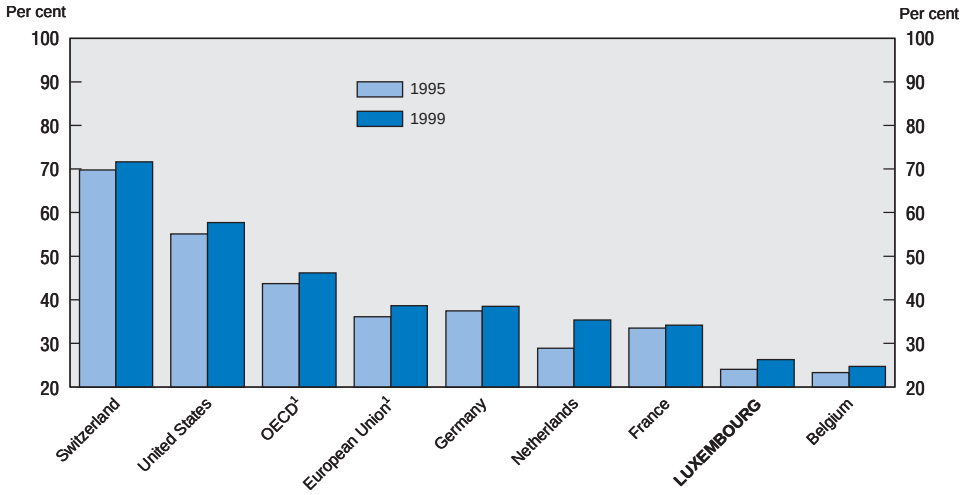
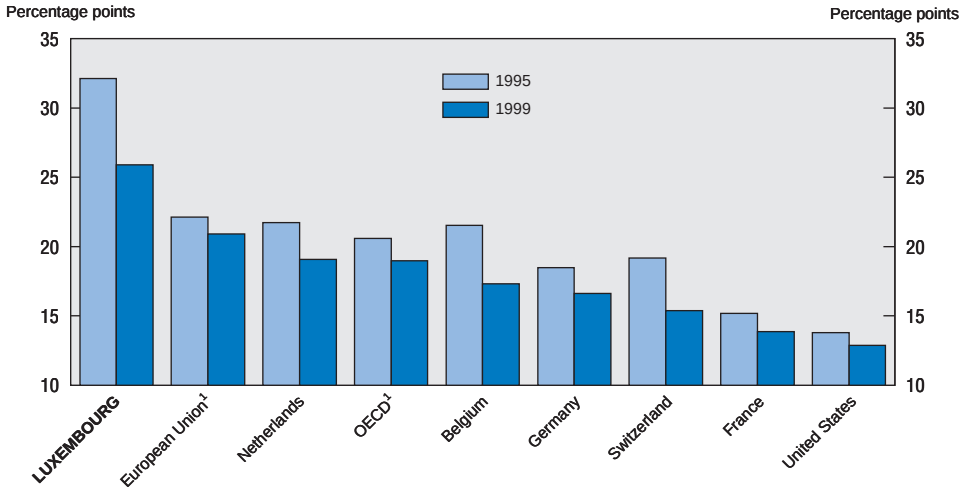
Figure 19. **Employment-population ratio for older age group, 55-64**



1. Simple average.
Source: OECD, *Labour Force Statistics*, Part III.

Figure 20. **The gender gap in employment-population ratios**

Difference between male and female employment ratios



1. Simple average.
Source: OECD, *Labour Force Statistics*, Part III.

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Table 6. **Working-age[1] persons receiving a benefit associated with non-employment**
Residents,[2] annual averages (thousands)

	1995	1997	1999
Disability	14.4	15.5	14.9
Early retirement (*Retraite anticipée*)	5.1	5.0	5.1
Pre-retirement(*Préretraite*)	1.1	1.0	0.9
Unemployment	2.6	3.2	2.2
Minimum social income (RMG)			
National solidarity fund (FNS)	3.7	4.1	4.6
Supplements from pension fund (*compléments caisse de pension*)[3]	1.6	1.6	1.3
Total	**28.5**	**30.4**	**29.0**
Resident population 15-64[4]	275	280	285
Total (as per cent of population 15-64)	**10.4**	**10.9**	**10.2**

1. 15-64.
2. Data for disability, early retirement and pre-retirement pensions are only available for residents and non-residents combined. However, the IGSS has estimated that the proportion of disability pensions paid to non-residents was, respectively, 28.1 per cent, 27.8 per cent and 28.8 per cent in 1995, 1997 and 1999. These proportions are used to estimate disability, early retirement and pre-retirement pensions paid to residents. Data for unemployment benefits and the RMG are for residents who alone are entitled to these benefits; cross-border workers are not entitled to them, despite paying social security contributions for unemployment insurance and contribution to the National Solidarity Fund, but rather are entitled to such benefits in their country of residence if it is an EU country.
3. This concerns people who receive a benefit from a pension scheme (invalidity or old-age) that is less than the RMG benefit [LUF 33 392 (€ 828) per month]. The pension scheme pays the difference between the RMG and the pension.
4. Labour Force Survey estimates.

Source: Inspection générale de la sécurité sociale (IGSS).

The low overall employment ratio is associated with a relatively high proportion of the working age population receiving social security benefits connected with not being employed (Table 6). By far the most important such benefit is disability pension, which has grown rapidly since its inception in the mid-1970s and now is estimated to account for about half of all such benefits. Early retirement pensions and the minimum social income (*Revenu Minimum Garanti*, RMG) account for most of the remaining such benefits. In all, an estimated 10 per cent of the working-age population receives a social security benefit associated with not being employed.

Older workers

Reasons for withdrawing from the labour force prematurely

The main reason that many older workers in Luxembourg withdraw from the labour force prematurely would appear to be that they prefer receiving generous social security benefits for not working. The most important such benefit is disability pension (Table 7). This may be granted to someone who is no longer apt to work in an occupation that corresponds to his or her strengths and aptitudes.[19] Representative disability pensions, which are calculated on the basis of a complete contributions

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Table 7. **Disability or early retirement pensions**
Residents[1] (aged 55-64), annual averages (thousands)

	1995	1997	1999
Disability	9.5	10.1	10.0
Early retirement (*Retraite anticipée*)	5.1	5.0	5.1
Pre-retirement (*Préretraite*)	1.1	1.0	0.9
Total	15.7	16.6	16.0
Resident populaton[2]	43.0	42.0	42.0
Estimated as a percentage of resident population	36.5	39.5	38.1

1. Data for disability, early retirement and pre-retirement pensions are only available for residents and non-residents combined. However, the IGSS has estimated that the proportion of disability pensions paid to non-residents was, respectively, 28.1 per cent, 27.8 per cent and 28.8 per cent in 1995, 1997 and 1999. These proportions are used to estimate disability, early retirement and pre-retirement pensions paid to residents.
2. Labour Force Survey estimates.

Source: Inspection générale de la sécurité sociale (IGGS).

record, range from 56 to 84 per cent of the average income subject to contributions on a gross basis (Table 8). Almost one quarter of the older working age population (aged 55-64) receives this benefit, a proportion that is far in excess of that (2 per cent in 1999) for the rest of the working age population. While disability undoubtedly increases with age, the extent of the deterioration apparent in the age distribution of benefit recipients is difficult to reconcile with data on observable medical ailments – on many scores, the health of the older working age population has improved relative to the rest of the population. However, older workers in many countries are finding the workplace ever more stressful and give this as a reason for stopping work.

The other main social security benefit that encourages premature withdrawal from the labour market is the early retirement pension (*retraite anticipée*). This pension is available after 40 years of actual contributions from age 57 or after 40 years of actual and implied contributions from age 60.[20] It is paid at the same rate as if the retiree had taken the pension at the official retirement age (65). Representative early retirement pensions are somewhat higher than disability pensions, ranging from 72 to 103 per cent of average incomes subject to contributions in gross terms and a little more in semi-net terms. Pre-retirement (*préretraite*) pensions are also available in industries that are experiencing economic difficulties.[21] These pensions allow employees to retire up to three years before qualifying for the early retirement pension. During this period, they receive 80, 75 and 70 per cent respectively of their last earnings. Approximately 14 per cent of older working age persons receive an early-retirement or a pre-retirement-pension, bringing to 38 per cent the total proportion of older working age persons receiving one of these benefits or a disability pension.

The decision to retire prematurely on a disability or early retirement pension is the employee's, not the employer's: employees taking early retirement by

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Table 8. **Representative[1] disability and retirement pensions, 1997**

		LUF[2]		Disability		Retirement	
				Per cent of average income subject to contributions			
		Disability	Retirement	Gross	Semi-net	Gross	Semi-net
AVI[3]	Men	65 803	80 700	84.0	..	103.0	112.0
	Women	40 881	45 208	69.9	..	77.3	84.0
CPEP[4]	Men	86 383	102 153	61.2	..	72.4	78.7
	Women	57 624	75 143	55.8	..	72.8	79.1
CPACI[5]	Men	65 325	79 395	66.2	..	80.4	87.4
	Women	44 460	51 552	66.9	..	77.6	84.3
CPA[6]	Men	40 081	42 138	75.7	..	79.6	86.5
	Women	37 639	39 727	75.7	..	79.9	86.8

1. "Representative" pensions are calculated on the basis of a complete contribution record (40 years for the retirement pension) and are not subject to any reduction resulting from the accumulation with other social security benefits.
2. LUF 40.3339 = € 1.
3. AVI (*L'établissement de l'assurance contre la vieillesse et l'invalidité pour les ouvriers*) = Establishment for old age and disability assurance for workers.
4. CPEP (*La caisse de pension des employés privés pour les employés privés et les travailleurs intellectuels indépendants*) = Pension system for private sector employees and self-employed intellectual workers.
5. CPACI (*La caisse de pension des artisans, des commerçants et industriels pour les professions indépendantes*) = Pension system for self-employed craft-workers, shop keepers and industrialists.
6. CPA (*La caisse de pension agricole pour les agriculteurs, les viticulteurs et les horticulteurs*) = Pension system for farmers, viticulturists and horticulturists.

Source: Inspection générale de la sécurité sociale (IGSS).

this means are expressing their own preferences, not necessarily those of their employers. By contrast, taking a pre-retirement pension (*préretraite*) requires the agreement of both the employer and the employee. The large number of older working age persons retiring prematurely on one or other of these pensions attests to the attractiveness of this option for many employees. Moreover, as noted by the Prime Minister, "…the number of those who, at 52-53 do all they can to stop working, continues to rise." (Speech on the Economic and Social State of the Nation, 10 May 2000.) At the same time, premature retirement of older workers also serves the interests of some employers, at least in cases where pre-retirement arrangements have been put in place. In the Prime Minister's view, "It is a true scandal that some businesses try to get rid of employees aged over 50 to replace them by younger, less expensive employees." (*Ibid.*)

Older workers are indeed more expensive than younger workers in Luxembourg (Table 9). Earnings continue to rise with age in relation to prime age (25-49) earnings for all major categories of education attainment at least until age 50-54. For the group with education attainment below upper secondary level, earnings peak in the 55-50 age group and for the group with education attainment at the

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Table 9. **Earnings progression by age and level of education attainment**

Age 25-49 = 100

	Below upper secondary			Upper secondary			Higher education			All education levels		
	Men	Women	Total	Men	Women	Total	Men	Women	Total	Men	Women	Total
40-44	109.6	110.1	109.7	113.9	118.2	114.9	129.0	131.2	129.4	111.8	112.6	111.9
45-49	113.0	104.9	111.8	130.5	119.6	128.5	140.6	136.4	140.2	117.2	108.6	115.9
50-54	120.5	107.1	118.7	132.2	139.4	133.1	149.0	136.4	148.5	123.9	111.5	122.4
55-59	124.8	116.7	123.7	143.6	139.1	143.2	139.7	..	140.0	128.8	118.8	127.6
60-64	122.3	95.2	114.1	162.9	142.7	157.5	132.6	..	130.0	131.7	104.7	124.4

Source: STATEC and OECD.

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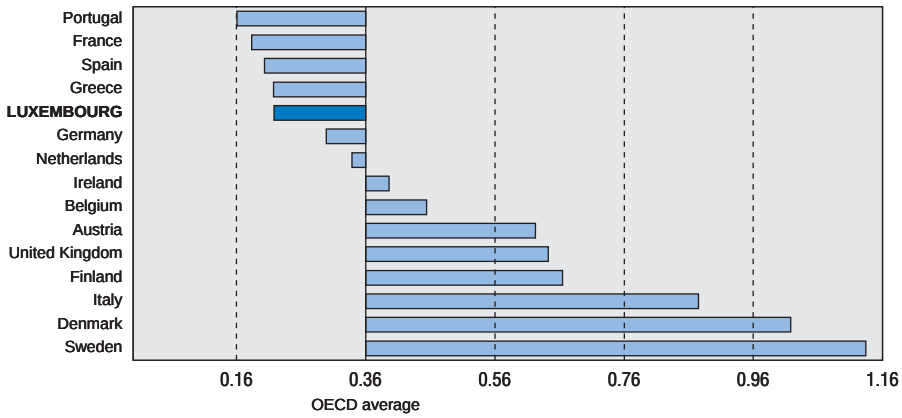
upper secondary level, earnings peak in the 60-64 age group.[22] This pattern contrasts with that in many OECD countries, where relative earnings (for a given level of education attainment) tend to decline beyond the age of 50 (OECD 1998*a*, pp. 132-34). Presumably, this reflects declining productivity relative to prime age workers. Traditionally, such a fall off in relative productivity has been associated with manual work, where older workers clearly no longer have the strength and stamina of younger workers. However, changes in the workplace in recent years associated with globalisation and the use of information technology appear to have exposed sedentary workers to this phenomenon. Job content now changes more quickly than in the past and often calls for a greater diversity of skills. In addition, most workers must now be able to use information technology effectively. These requirements place older workers at a disadvantage, as they tend to have greater difficulty in learning new skills and adapting to rapidly changing circumstances. In countries (such as Luxembourg) where institutional and social arrangements prevent older workers from compensating for a decline in relative productivity by lower relative earnings, such workers may tend to experience greater employment difficulties (*ibid*).

Older workers' disadvantage on the labour market is reinforced by the tendency for them to receive less continuing education than other workers do (Figure 21). This tendency is particularly marked in Luxembourg, where the ratio

Figure 21. **Career or Job-related training by age, 1997**

Ratio of participation of older workers to younger workers[1]



1. Younger is defined as 25-29 and older as 50-54. These data come from European Labour Force Survey (ELFS). Training is defined as education and training received for a reason other than secondary education or initial vocational training.

Source: Eurostat (ELFS), 1997.

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of participation in work-related training of older workers (50-54) to younger workers (25-29) is one of the lowest in Europe. A major reason for less being invested in the skills of older workers is that the investment must be amortised over a shorter period. This factor is especially important in Luxembourg owing to the low effective retirement age (Figure 22). Another consideration in Luxembourg is that there is a very large pool of skilled workers in the surrounding regions, which could reduce the pressure on employers to train existing employees: it may be cheaper to hire workers who already have the required skills. Finally, investment in training is often lower for older workers because they have a lower level of educational attainment, reducing the returns on such investment.

Measures to increase labour force participation of older workers

The government is committed under the (EU) National Action Plan to Promote Employment (NAP) to reducing the risk of older workers sliding out of the labour force into one of the passive programmes (disability, early retirement and RMG) that provide support for the economically inactive. The law of 12 February 1999 on the NAP modified the early retirement system by creating the possibility of a progressive early retirement[23] (as from 57) or of a progressive shift to part-time work[24] (workers aged over 49), both of which are subsidised. In either case compensatory recruitment is required. Furthermore, the government plans to offer *training* specifically aimed at older workers (aged 50 or more) in the use of information technology. This training, which will be in addition to the measures planned for all age groups (greater efforts at orientation, individually-adapted training and systematic follow-up), is intended to provide key-qualifications to enable older workers to participate fully in the information society. Such training is to be offered to employed persons, reducing the risk that they become unemployed, as well as to older unemployed persons, enhancing their prospects of finding a new job. Moreover, the government, in consultation with the social partners, plans to reform the law on *disability pensions*. The government intends to provide *assistance* for older workers who are no longer as productive as before so that their work conditions can be adapted to their specific needs. Financial aid will be made available to employees (compensation for possible loss of income) and to employers participating in this scheme to reintegrate older workers in their current firm. In addition, the government has embarked on an awareness-raising campaign to dissuade people from retiring early on a disability-, early- or pre-retirement pension. Although the government considers that retirement at 57 should not become the social norm, no measures are planned to increase the age at which older workers can take an early retirement pension (*rétraite anticipée*). Moreover, there is still pressure from certain business sectors to retain the pre-retirement (*préretaite*) arrangements that facilitate an even earlier retirement.[25]

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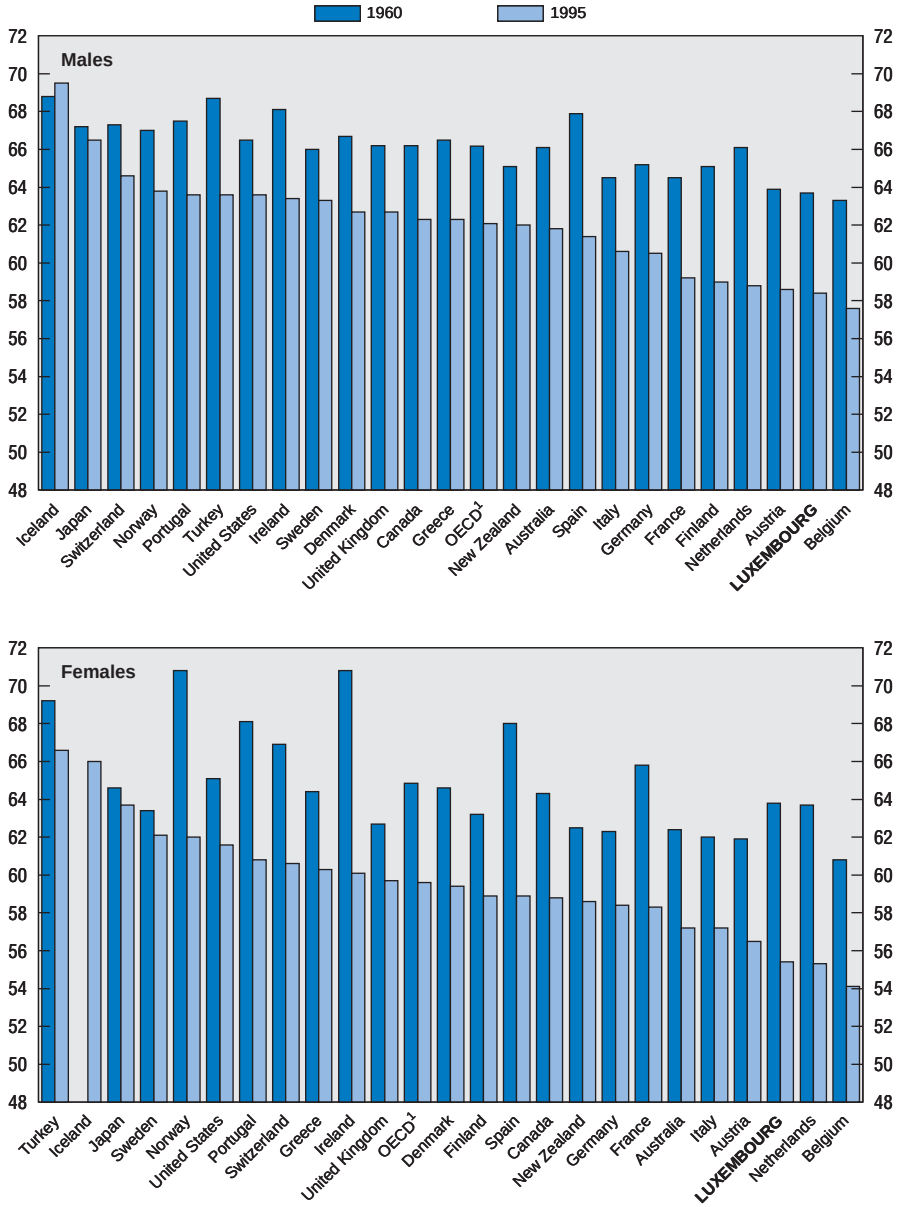
Figure 22. **Estimates of the average age of retirement**



1. Unweighted average.
Source: ILO, *Economically Active Population 1950-2010*, December 1996.

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Insofar as the low labour force participation of older workers reflects voluntary withdrawal from the labour force, restricting access to disability benefit will be a critical step in increasing participation rates. While the measures to encourage progressive retirement should also help, further steps are likely to be required to discourage early retirement on the early retirement pension and the pre-retirement pension. Given that the early retirement pension is the same as the pension taken at the official retirement age, older workers have little financial incentive to continue working once they qualify for an early retirement pension. One way of enhancing the incentive to continue working would be to make an actuarial downward adjustment to pensions taken before the official retirement age to reflect the longer period during which they can be expected to be drawn. An equivalent upward adjustment could be made to pensions drawn after the official retirement age. Alternatively, the age at which people qualify for early retirement could be raised. Government subsidies for pre-retirement pensions should also be terminated. These subsidies are based on the "lump of labour" theory according to which there is only a certain amount of work to go around. On this basis, measures that prematurely eject older workers from the labour force or in other ways reduce the supply of labour (such as restricting the working week to 35 hours) are thought to reduce (structural) unemployment. In fact, there is no empirical support whatsoever for this view.[26] Subsidies for before retirement arrangements simply reduce the employment-to-population ratio and subsidise declining industries at the expense of other industries. This slows industrial adjustment, to the detriment of productivity growth in the overall economy. Finding ways to help older workers to cope better with the pressures of modern working life, such as the planned training in the use of information technology, should also help to reduce premature retirement.

In the event that premature voluntary withdrawal from the labour force does decline, it may be necessary to relax employment protection regulation (EPR)[27] to ensure that firms maintain adequate flexibility to adjust employment to adverse demand shocks and to reduce the reluctance of employers to hire older job seekers.[28] Particularly important in this regard would be some relaxation in Luxembourg's strict rules on dismissals and the use of fixed-term (including temporary) contracts. An indication of the strictness of these rules in relation to hiring is that some 95 per cent of employment contracts[29] are for an indefinite term, the highest proportion in the EU (Figure 23). Temporary employment amounts to less than 2 per cent of total employment, with the remaining 3 per cent being permanent employees on fixed-term contracts. Less strict rules on dismissals and the use of fixed-term contracts would, however, increase the number of older workers who are laid-off by their employers. As this group typically experiences great difficulty in the labour market, such persons would require considerable assistance with finding another job. This underlines the importance of the measures planned to increase the employability of older workers.

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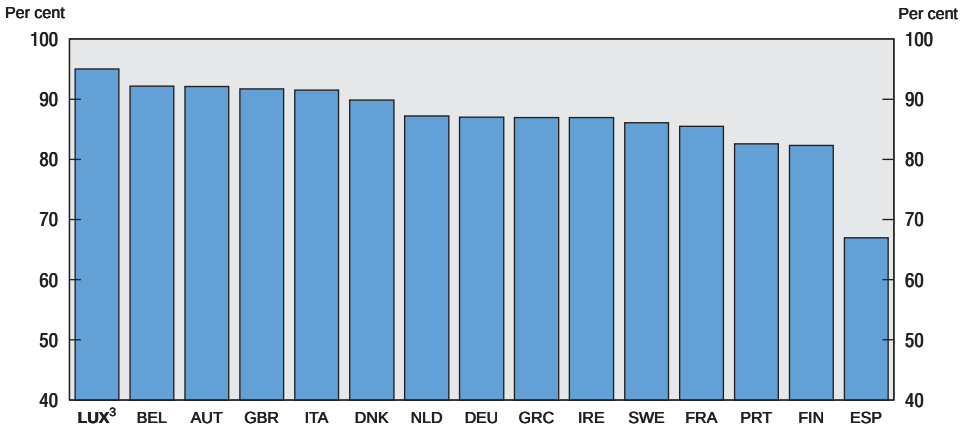
Figure 23. **The proportion of indefinite term employment contracts,[1] 1998**
As a percentage of total employees[2]



1. Nace Rev.1, total.
2. Army included.
3. Domestic concept.

Source: Eurostat, *Population and social conditions*.

The large gender gap in economic activity rates

Causes

Strict employment protection regulation, as in Luxembourg, tends to be associated with a high employment ratio for prime age males and low ratios for each of the other demographic groups, including women, both of prime age and of older working age (OECD 1999, *Employment Outlook*). Strict regulation of part-time employment[30] contracts in Luxembourg further weakens female activity rates. Such regulation discourages employers from offering part-time contracts: the proportion of part-time employment in total employment in Luxembourg is considerably lower than in most other countries, although it has been increasing in recent years (Figure 24). This situation mostly harms women as they overwhelmingly fill part-time jobs: part-time employment is an important means by which many women are able to reconcile family and working life. Inadequate child-care facilities in Luxembourg also restrain female activity rates. As an indication of the difficulties that women have in reconciling family and work life, parenthood has a greater adverse effect on the gender gap in employment in Luxembourg than in most other European countries[31] (Table 10). Another factor which may contribute to the gender gap in employment is that the basic unit for personal income tax is

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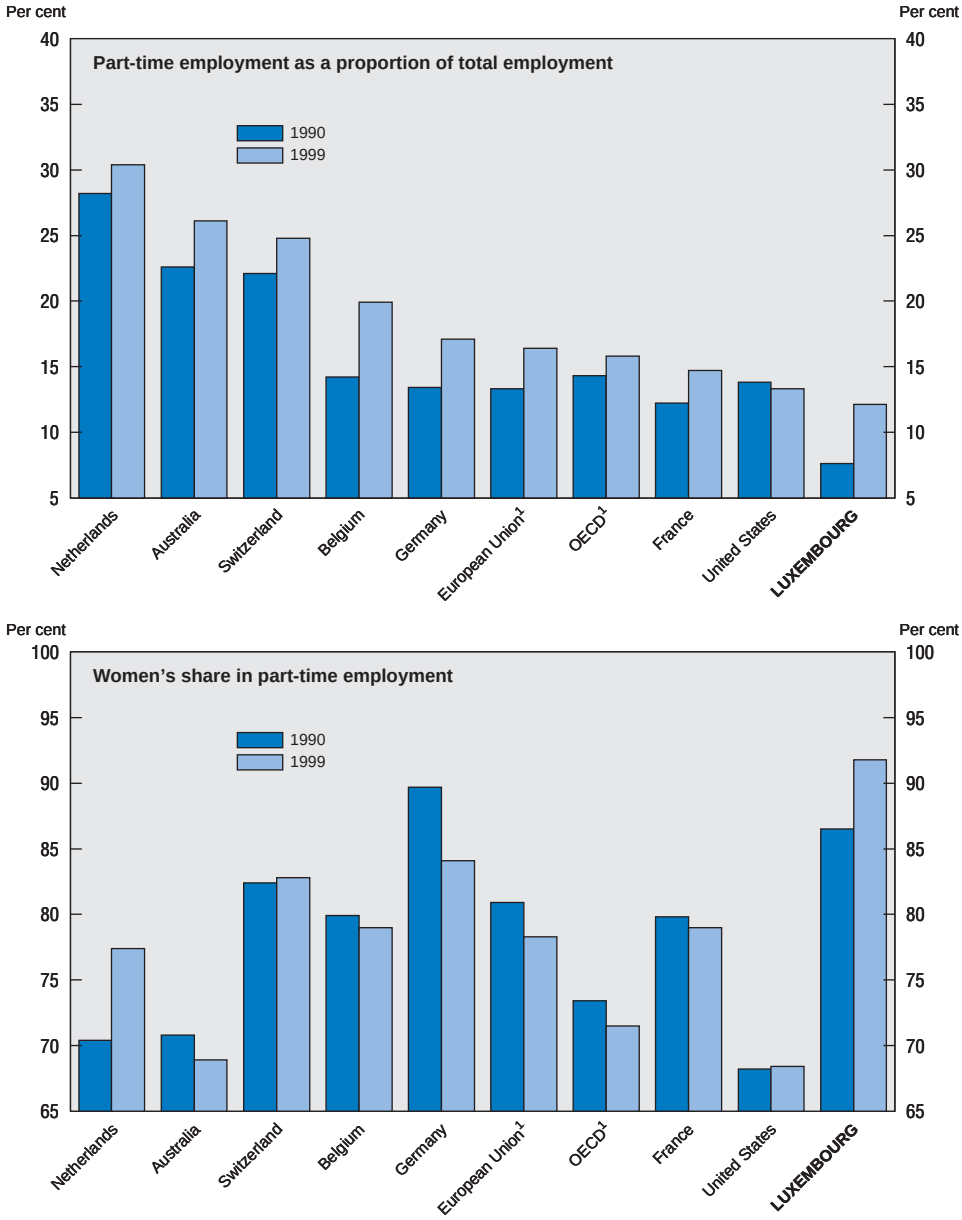
Figure 24. **Incidence and composition of part-time employment**



1. Simple average.
Source: OECD, *Employment Outlook*, June 2000.

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Table 10. **Influence of parenthood on employment by gender**

	Comparative index[2]	Employment rates,[1] 1998			
		No children		Children aged 0-5	
		Men	Women	Men	Women
Ireland (1997)	1.9	81	81	87	46
Luxembourg	**1.8**	**93**	**84**	**98**	**49**
United Kingdom	1.7	85	87	90	53
Spain	1.7	88	67	90	40
EU11	1.7	84	79	91	52
Netherlands	1.6	88	86	95	60
Italy	1.6	91	68	92	45
France	1.5	79	72	91	56
Germany	1.5	82	60	96	48
Austria	1.3	89	85	94	68
Belgium	1.2	87	77	92	66
Portugal	1.2	92	83	96	72

1. 20 to 44 year olds.
2. This compares employment rates by gender for persons without a child with those for persons with a child aged less than 5. The closer the value of the index to one, the more similar the impact of parenthood on the employment rates of women and men. Values over one suggest that the impact of parenthood on females has reduced the chance of employment relative to the experience of males. Data for Sweden, Denmark and Finland are not available.

Source: European Union (1999), *Joint Employment Report* 1999, Part I.

the household rather than the individual. This means that women frequently face high marginal tax rates, especially if their husbands have high earnings. Other factors, cited by the authorities, are strong traditional role models that are reinforced in school and by the social partners and high salaries, which reduce the financial need for wives to work.

Measures to increase female activity rates

The government has taken a number of measures in the context of its NAP to increase female activity rates. To reinforce equal opportunities policies, it has introduced professional training addressed exclusively to women[32] and subsidies to private businesses that re-organise working arrangements to facilitate the reconciliation of private and professional lives.[33] In order to reduce discrimination, measures have been taken to encourage females to choose scientific and technical subjects in their education and a legal basis has been introduced for positive discrimination in private businesses. Dispositions have also been taken to pay subsidies to employers hiring women in certain circumstances[34] in professions in which women are underrepresented; these professions have yet to be determined. To enable parents more easily to reconcile professional and family life, the government has committed to creating more places in day-care centres for chil-

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dren and has introduced parental leave[35] and leave for family reasons. In addition, part-time employment contracts have been made more flexible,[36] which should encourage their development. Finally, the reintegration of women into working life will be facilitated by the new law on continuing education which makes such education available to persons on leave or temporarily absent from their employers.

The Prime Minister recently announced plans for a substantial increase in the number of places available in childcare centres, both at the pre-school level (for children aged 0-4) and for younger school children (aged 5-13) (Speech on the Economic, Social and Financial State of the Nation, 10 May 2000). The government intends to increase the number of places available for pre-school children by 30 per cent and for school aged children by between 100 and 160 per cent. This will bring the number of places available to 17 per cent of the population aged 0-4 and to between 16 per cent and 21 per cent of the population aged 5-13. There will be fees for using these facilities that will depend on parents' income. Parents with high incomes will not be subsidised. The government also intends to undertake a study on the systems for pensions, social transfers and taxation that will take into account the effects of these systems on female participation in the labour force.

For the time being, no further liberalisation of part-time employment contracts, which mostly concern women, is planned. Workers' representatives in the Economic and Social Council (*Conseil économique et social*, CES) consider that this is appropriate: in their view, it would be best to wait and see the results of the recent reform in part-time employment contracts contained in the law of 12 February 1999 on the NAP before taking further action. Employers' representatives, on the other hand, suggest that the government should take advantage of the Community Directive on part-time employment contracts to redefine this kind of contract with a view to removing the rigidities that tend to dissuade employers from offering part-time jobs. Overall, the consensus of CES members is that the current legislation on part-time employment still does not take enough account of the needs of workers and employers and that greater use of part-time contracts is likely to be an important factor in raising female activity rates (CES, p. 81). (It would also facilitate a progressive departure of older workers from working life.) Some relaxation in regulations concerning temporary employment and employment protection could also help to increase female activity rates. In this connection, lowering barriers to the development of tele-working could be particularly beneficial for women.

Other labour market reforms

More flexible working time

The law of 12 February 1999 on the NAP provides for greater flexibility in working time. Henceforth, the reference period over which working time is mea-

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sured may be up to one year, compared with the former one week, subject to the constraint that the average working week over a four-week period does not exceed the standard working week (presently 40 hours). As in other countries, this change responds to the needs of businesses for greater flexibility in production schedules. It should increase productivity, resulting in higher real wage rates. This may not, however, lead to a very large increase in labour force participation in Luxembourg because the main social security benefits that provide replacement income for people not working are indexed to wage levels. The increased flexibility in working time provided for by the law has not yet been implemented in collective agreements owing to a disagreement between employers and unions about whether the reference period can be less than four weeks: unions argue that it can but employers disagree.

The social partners are also negotiating a voluntary (*i.e.* without government coercion) reduction in working time. This too is unlikely to have much effect on the employment ratio. Even if there were little change in real product wage rates (*i.e.* real wage rates adjusted for labour productivity), the increased hourly wage rates being compensated by increased productivity, most new jobs would be likely to go to cross-border workers. This is because Luxembourg residents not working or considering withdrawing from the labour force would have no greater incentive to work owing to the fact that, as noted above, the major benefits providing replacement incomes are linked to wages. In the event that real product wage rates rose as a result of the move to a shorter working week, this could be expected to reduce total demand for labour (hours), reducing employment opportunities mainly for cross-border workers.

Developing lifelong learning

Some studies have found that the recruitment of cross-border workers is partly explained by their level of training, their flexibility, their geographical mobility and their adaptability (EU 2000*a*, *Joint Employment Report*, p. 143).[37] In view of this, Luxembourg's NAP aims to increase the proportion of employment growth going to residents by improving their skills. To this end, a new framework law on lifelong learning came into effect in January 2000. It provides tax reductions for enterprise-based training. This reform should help to raise the amount of job-related training in Luxembourg, which is low by international comparison (Figure 25). As discussed above, the low level of such training may partly be caused by the ease with which employers can hire workers in the surrounding region who already have the required skills.

A number of measures have also been taken or are planned to improve the performance of the school system. This is intended to provide a stronger foundation for the whole system of lifelong learning; as noted above, the amount of job-related training received tends to be positively related to education attainment,

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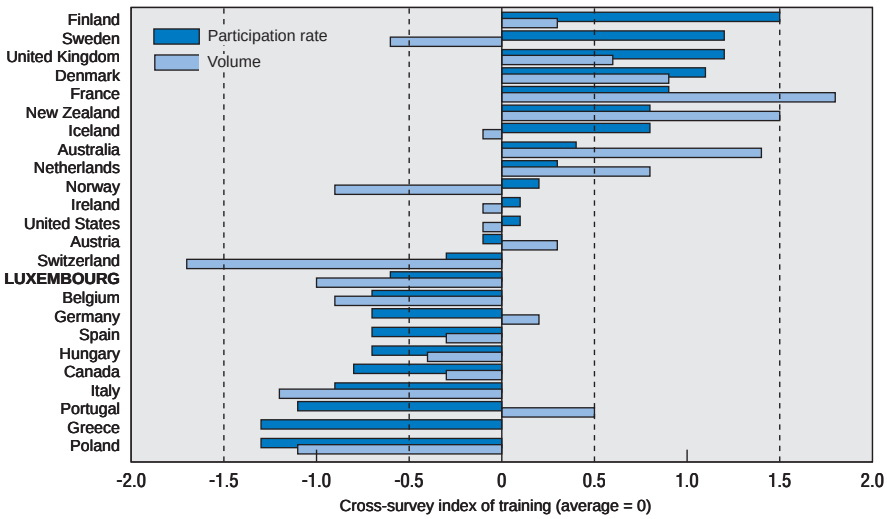
Figure 25. **Job-related training: participation and volume**[1]



1. Cross-survey indices. Such training is defined as education and training received for a reason other than secondary education or initial vocational training.

Source: OECD, *Employment Outlook*, June 1999.

presumably because the returns on training people who are already well educated tend to be higher. The school curriculum is being refocused on core knowledge, leaving more time for students to put into practice what they learn and thereby gain key competencies. Moreover, relations with the social partners have been improved, with agreement being reached that school education should aim to provide all-round education pertinent to working life as well as preparation for citizenship and personal development. A methodology has also been developed with co-operation from business groups that has the objective of defining and adapting vocational training curricula to the needs of the workplace. A number of measures are also being considered to improve the quality of school education. One project that could be particularly important for increasing the proportion of students who complete at least upper secondary level studies (estimated to be 70-74 per cent, compared with an OECD average of 79 per cent in 1998) would reduce the linguistic barriers to education for immigrants posed by instruction in Luxembourg's three languages. There are also projects to improve teaching methods and teacher training, better integrate information technology, and to increase the autonomy of both students and schools.

Access to apprenticeships is also to be facilitated through the introduction in the 2000-01 academic year of a modular approach to training. In addition,

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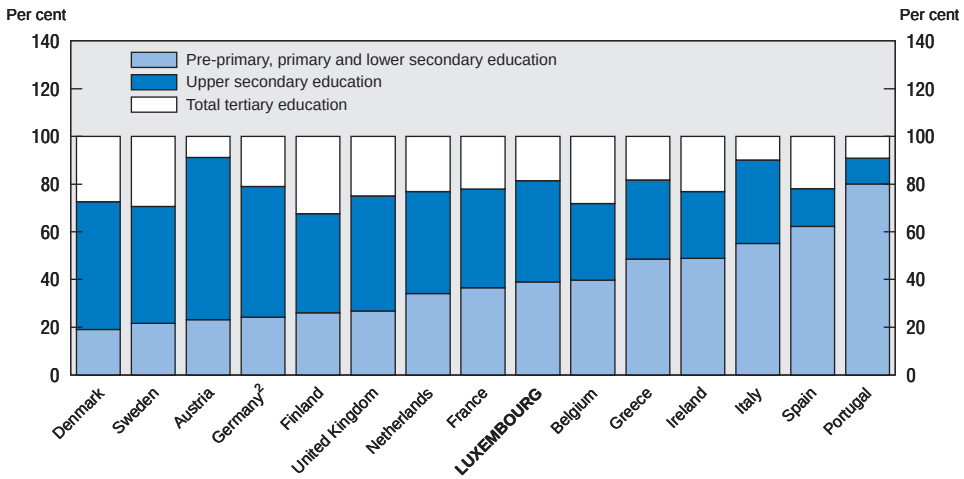
certain groups of apprentices will receive an income equivalent to the minimum social income (RMG), encouraging them to complete their training rather than to opt for economic inactivity. Employers have also substantially increased the number of apprenticeship places on offer, although many of these are not taken up because of the lack of resident applicants.

Measures are also being taken or are planned to increase the proportion of the population completing studies at the higher education level: at any given time, only 1.7 per cent of the population is undertaking such studies, compared with a European average of 2.5 per cent. The loan system for such studies has been modified to include a wider field of studies and incentives have been introduced to encourage students to finish their studies on time. These loans are particularly important in Luxembourg as most such studies are undertaken abroad. Planning is also proceeding for the creation of institutes of advanced studies. These institutes, which will combine research and teaching at post-graduate level, are designed to complement areas of excellence to be found in various economic sectors (such as the financial and communications sectors). In all, these measures should help to increase the level of educational attainment of the adult population, which is rather low by international comparison, even abstracting from immigrants (Figure 26).

Figure 26. **Education attainment of the population aged 25-59, 1999**[1]



1. 1997 for Ireland and 1998 for Greece.
2. Western Germany.

Source: Eurostat, *Population and social conditions*.

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Other barriers to labour market flexibility

As noted above, Luxembourg has been spared the high unemployment experienced by other European countries with similar labour market institutions because favourable shocks, notably the development of the financial and communications sectors, have outweighed the negative shocks experienced by all countries. However, like any small regional economy, the degree of specialisation required to achieve high living standards renders the economy vulnerable to developments in the key sectors. Just as this can drive the economy to very high growth rates, as in recent years, adverse developments in these sectors could have a severe negative effect on the economy.[38] Reforms to make labour markets more flexible would help Luxembourg to cope with such adverse shocks were they to occur. The main institutional features of the labour market that weaken the capacity to adapt to adverse shocks have been documented in past OECD *Surveys* and are summarised in Box 2 (Follow-up on OECD recommendations for structural reform since 1998). Some of them, notably restrictive employment protection regulation and impediments to the use of part-time contracts, have already been discussed. In the remainder of this section, some of the others that could be particularly damaging to economic performance in the event of an adverse shock are discussed.

Wage indexation

Wages, pensions and other social security benefits are indexed to consumer prices.[39] They are adjusted upwards once cumulative inflation reaches 2.5 percentage points since the last index-linked increase. As noted in Chapter I, the energy price shock has sharply shortened the intervals between indexed wage increases, with the latest increase (in July 1999) occurring just 11 months after the previous adjustment.

The problem with wage indexation is that it could hinder adjustment of real wages to an adverse supply shock, such as an increase in energy prices. Were this to occur, the ensuing rise in real product wages (nominal wages deflated by the GDP deflator) would lower profitability and reduce demand for labour. While this would be likely to fall disproportionately on cross-border workers and net inflows of immigrants, it is nevertheless probable that there would be an adverse effect on national employment and the unemployment rate. At the same time, such a shock could set off a domestic wage-price spiral, with shorter intervals between indexed wage increases. Such a spiral would further undermine national competitiveness and reduce demand for labour still further. The wage indexation law of 1975 provides a means of breaking such a vicious circle in that automatic indexation can be suspended in difficult economic circumstances, as occurred at the time of the devaluation of the Luxembourg franc in the early 1980s. However,

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Box 2. Follow-up on OECD recommendations for structural reform since 1998

Proposal	Actions	Assessment/recommendations
Labour markets		
1. Reform early retirement- and disability pensions		
• Tighten access to early retirement and disability schemes. Re-examine beneficiaries at regular intervals and promote their integration into the labour market.	The 1999 law on the NAP created the possibility of a progressive early retirement (as from 57) or of a progressive shift to part-time work (for workers aged over 49). The government plans to: • offer training specifically aimed at older workers in the use of information technology; • reform the law on disability pensions; and • provide financial assistance for older workers who are no longer as productive as before so that they can remain in employment.	Further measures would be helpful for raising the employment ratio for older workers. Incentives to continue working should be enhanced. These could take the form of lower income replacement rates on disability and early retirement pensions. In the case of early retirement pensions, they could be reduced actuarially as a function of the age at which they are taken. The counterpart could be an actuarially based increase in pensions taken after the official retirement age. Alternatively, the age at which an early retirement pension can be taken could be increased. Pre-retirement pensions (*préretraite*) should be phased out. Review the effectiveness of the new law on disability pensions as soon as a long enough period to make this assessment has passed, and, in the event that there has not been a substantial decline in the number of invalidity pensioners, take further steps to discourage recourse to this pension.
2. Reform unemployment and related benefit schemes		
• Strengthen incentives to look for work or a place in a training scheme at least by intensifying surveillance on being available for work and job search and by applying more strictly sanctions for non-compliance with requirements for receiving unemployment benefit.	The 1999 law on the NAP intensified surveillance on availability for work and sanctions for non-compliance.	No further action is necessary. The low rate of unemployment suggests that this is not a serious problem.

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Box 2. **Follow-up on OECD recommendations for structural reform since 1998** (*cont.*)

Proposal	Actions	Assessment/recommendations
• Reduce the withdrawal rate of benefits for additional earnings under the general assistance scheme (RMG), including housing assistance.	None.	The recommendation is maintained.
3. Increase employment flexibility		
• Continue to reform the strict working hours legislation and remove obstacles for part-time work and temporary work contracts.	The 1999 law on the NAP provides: • for a reference period over which working time is measured of up to one year, subject to the condition that the average working week over a four-week period does not exceed the standard working week; • that, unless excluded by the employment contract, part-time workers may work up to 20 per cent longer than the daily or weekly duration provided for in the contract without being paid overtime rates subject to the condition that the average working week over four consecutive weeks does not exceed the normal working week fixed in the contract.	• Remove the remaining rigidities that dissuade employers from offering part-time jobs. • Relax regulations on temporary employment. • Make rules on dismissals and the use of fixed-term contracts less strict.
	The government plans to create more places in day-care centres.	This should help parents to reconcile family and work responsibilities, thereby facilitating an increase in the female employment ratio.
4. Increase wage and labour cost flexibility		
• Improve flexibility in wage bargaining, notably by further limiting the administrative extension of sector wage agreements, and phase out wage indexation or at least exclude energy from the index.	None.	Wage indexation should be modified to exclude energy from the index. Wage agreements should allow for lower wages for older workers whose productivity declines so as to enable them to remain in employment.

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Box 2. **Follow-up on OECD recommendations for structural reform since 1998** (*cont.*)

Proposal	Actions	Assessment/recommendations
• Reduce the impact of the minimum wage by either reducing the minimum wage for certain groups, such as young people above the age of 17, in combination with in-work benefits or by subsidies to employers.	None.	The government does not consider that such actions would be useful. Nevertheless, a shift to reliance on a combination of minimum wages and targeted in-work benefits for income redistribution objectives could contribute to greater employment of low-skilled or inexperienced workers.
• If for political or social reasons wage flexibility cannot be increased sufficiently, further reduce taxes and employers' social security contributions for low-wage earners or at least for blue-collar workers.	The income tax reform in 2001-02 gives the largest proportionate tax cuts to low income earners.	The low rate of unemployment suggests that labour costs are not a major barrier to employment, even of the low skilled. A move from the household to the individual as the relevant unit for income tax should be considered as a means of raising the low female employment ratio.
5. Expand and enhance active labour market policies		
• Apply more strictly the working requirement in the RMG scheme.	The public employment service has been remodelled and activation measures have been strengthened (see below).	Further efforts are required to get people on the RMG back to work. Lower benefit withdrawal rates could encourage beneficiaries to return to work progressively, by working part-time.
	• Luxembourg set itself the objective of providing activation measures to all young persons within three months of becoming unemployed and to adults within six months of becoming unemployed. Non-compliance with these objectives fell from 29.9 per cent in 1998 to 13.8 per cent in 1999 for young people while the same rate for adults improved from 63.1 per cent to 41.4 per cent over the same period.	Continue to improve orientation, training and follow-up of unemployed persons.
	• Greater efforts at orientation, individually adapted training and systematic follow-up of unemployed persons are planned.	

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Box 2. **Follow-up on OECD recommendations for structural reform since 1998** (*cont.*)

Proposal	Actions	Assessment/recommendations
6. Promote lifelong learning	• A new framework law on lifelong learning came into effect in January 2000. It provides tax reductions for enterprise-based training. • The school curriculum has been refocused on key competencies. • A methodology has been developed with co-operation from business groups for defining and adapting vocational education to the needs of the workplace. • Measures are being considered to reduce linguistic barriers to education for immigrants. • Access to apprenticeships is to be improved and some apprentices are to receive an income equivalent to the social minimum (RMG). • The loan system for higher education has been modified to cover a wider range of subjects and to encourage students to finish their studies on time. • Continuing education focused on women has been introduced. As noted above, continuing education focused on older workers is planned.	Continue to build an effective system of lifelong learning on the foundations that have been established.
Product markets	• The telecommunications "last mile" was opened to competition in January 2001. • A law has been passed on the progressive opening of the electricity market to competition over 2001-05. • A new law is planned on liberalising the gas market to bring Luxembourg into conformity with Single Market commitments. • A law has been passed on the partial opening of the domestic postal market to competition. • A law on electronic signatures was passed in July 2000.	• Maintain progress in implementing EU single market legislation in a timely manner. • Continue to develop the legal framework necessary to support the development of electronic commerce.

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Box 2. **Follow-up on OECD recommendations for structural reform since 1998** (*cont.*)

Proposal	Actions	Assessment/recommendations
Reduce the costs of moral hazard in health care		
• Consider requiring general practitioners to act as gatekeepers for access to specialists and hospital care.	• The introduction of the gatekeeper system is not being discussed actively at the present moment. Nevertheless the general practitioners' professional organisation has proposed a very weakened form of gatekeeper system in which the general practitioner (GP) (would centralise the medical files of his patients.	Support the proposal for GPs to centralise their patients' medical files.
• Consider changing remuneration of medical practitioners to reduce the fee-for-service component.	• At the present moment there are no discussions about changing the remuneration system of medical practitioners.	The recommendation is maintained.
• Consider increasing co-payments.	• For 2000 co-payments have been increased as a means of reducing the expenditures of the health care insurance.	• In increasing co-payments, the benefits of reducing the costs of moral hazard need to be weighed against the costs of declining equity in access to health care, as has been done by the authorities.
• Consider benchmarking hospitals and introducing lump-sum payments based on the Diagnostic Related Groups-based system.	• The law concerning the health insurance allows the negotiation between the health insurance scheme and the "Entente des hôpitaux" of the remuneration of hospitals by DRG groups. For the time being, no such negotiation is being considered. It has to be stressed that the information system and analytical accounting system of the hospitals has still to be further developed to allow the introduction of DRGs.	• Develop the information and accounting system of the hospitals so that DRGs can be introduced.

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Box 2. **Follow-up on OECD recommendations for structural reform since 1998** (*cont.*)

Proposal	Actions	Assessment/recommendations
• The Ministry of Social Affairs and the health fund (UCM) need to be more closely involved in major investment decisions in the hospital sector.	• At the present time no changes are considered concerning the rules governing major investment decisions in the hospital sector. Concerning hospitals, a new hospital plan is presently under discussion.	
• Consider revising the tripartite structure of the UCM or at least the role of the social partners so that the same negotiating parties are not on both sides of the table.	• The programme of the government considers the revision of the administrative structure of the UCM during the next four years.	

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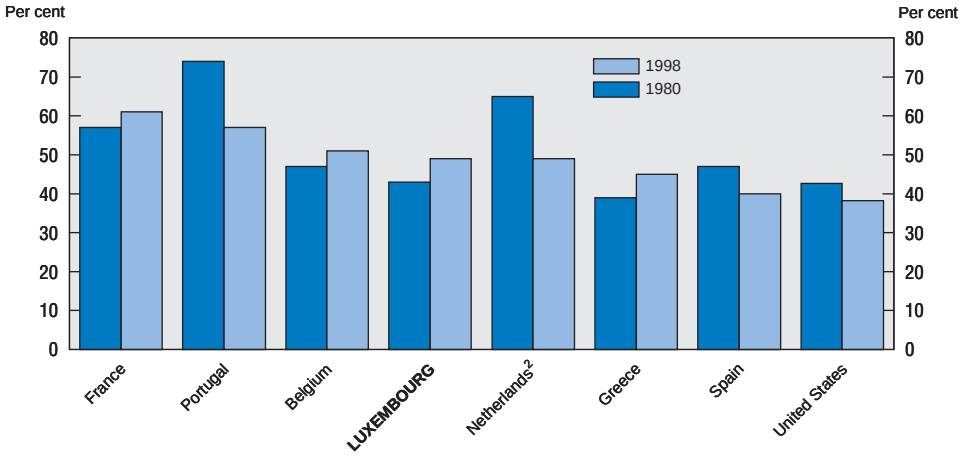
in view of the exceptional circumstances required to suspend indexation, it would be preferable to exclude energy prices from the relevant index (as in Belgium).

Minimum wage

Luxembourg is one of 17 OECD countries that has a statutory or national minimum wage, the *Salaire social minimum*, which cuts across most sectors of the economy (OECD, 1998). It is set by statute, only covers the private sector and is indexed. In addition, it is reviewed biennially in line with economic and pay growth. This review explicity takes into account the "expected" impact on employment, unemployment and competitiveness (*ibid*). The minimum wage rate amounts to a little over half of male average earnings in manufacturing, compared with around 40-60 per cent in the other countries for which data are available (Figure 27). Minimum wage rates for young persons (15-17 years old) are 60-80 per cent of the adult rate.[40] The minimum wage rate has increased somewhat relative to earnings since 1980, as it has in about half of the other countries for which data are available.

Minimum wage rates can reduce demand for low-skilled workers, increasing unemployment amongst this group. Their effect depends on how high they

Figure 27. **Minimum monthly wages as a proportion of average earnings**[1]
For single male manual workers in manufacturing industry



1. Gross earnings; for United States: hourly wages.
2. 1997 instead of 1998.
Source: Eurostat, *Population and social conditions* and OECD, *Main Economic Indicators*.

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are; a very low minimum wage would have little or no effect on actual wage rates, and hence on employment and unemployment. An indication of the extent to which the minimum wage constrains wage fixing is given by the proportion of workers paid at or near this wage rate. This proportion was 16.8 per cent in 1999 (*Conseil économique et social*, 2000), which is high by international comparison.[41] The proportion of workers paid the minimum wage has increased markedly in recent years, from 12.5 per cent in 1988 (*ibid*), reflecting a structural increase in the minimum wage rate. According to the Economic and Social Council (*Conseil économique et social*), outsourcing by the government sector, the railways and banks has also contributed to the increase in the proportion of workers paid the minimum wage. A greater proportion of women (22 per cent in 1999) is paid at around the minimum wage rate than men (14 per cent), although this difference is narrowing. This suggests that the minimum wage may have a more adverse effect on employment and unemployment for females than for males.

Minimum wage rates are intended to reduce inequality in the distribution of income.[42] They clearly benefit low-income persons who remain in employment and are paid around the minimum wage rate. However, their impact on household poverty and income distribution is limited because many poor households have no one working and many minimum-wage workers live in households with above-average incomes. Moreover, minimum wage rates may make some poor households even worse off by pricing members out of a job altogether. An alternative to minimum wage rates for reducing income inequality is to provide targeted in-work benefits (including payroll tax reductions on low wages). This approach is relied upon in many OECD countries, but not in Luxembourg. Its main advantage is that it does not price low-skilled persons out of jobs. The main disadvantages are that it may give rise to poverty traps, may lead to a fall in wages for low-skilled workers and may be fiscally expensive. In view of the advantages and disadvantages of the two approaches, there may be merit in placing less reliance on the minimum wage rate and introducing compensating in-work benefits.

High replacement rates on social security benefits

Luxembourg has very generous social security benefits. As discussed above, the disability and early retirement pensions give beneficiaries replacement rates that are frequently over 70 per cent and in some cases are over 100 per cent on a net basis. Unemployment benefit is also generous. It is 80 per cent of former gross earnings, which corresponds to an even higher net replacement rate when family and housing benefits are taken into account.[43] Although unemployment benefit is limited to one year, it is followed by generous social assistance. This would still give a low-paid worker (on two-thirds of the average production worker's wage) a net replacement rate of over 80 per cent after five years. Social assistance recipients are discouraged from taking part-time employment by a

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benefit withdrawal rate of almost 100 per cent. These arrangements result in high reservation wages and discourage social security beneficiaries from moving progressively back into employment. International experience suggests that high benefit replacement rates and long duration of benefits are two of the most important factors causing sustained increases in unemployment when economies are shaken by adverse supply shocks (Blanchard and Wolfers, 1999). These arrangements should be reviewed so that Luxembourg will be better prepared to weather adverse shocks when they arrive.

Product markets

Implementing EU *Single Market directives*

EU Single Market directives shape reform of product market regulation in Luxembourg, as in other EU countries. Implementing these directives has entailed substantial reforms in major network industries in recent years. In general, this has involved separating natural monopoly elements from potentially competitive elements and creating a regulatory environment conducive to competition in the latter. Of the major network industries, most progress has been made in liberalising *telecommunications*. This industry was opened to competition in July 1998 with the exception of the "last mile" and an independent regulator was established. In the year following liberalisation, prices of international calls fell by 35 per cent. They have continued falling since then, and are now 70 per cent lower than when the reform was announced in 1997. The number of telecommunications companies operating in Luxembourg has risen sharply over this period, to 19. On a purchasing power parity (PPP) basis, charges for a basket of services have fallen to amongst the lowest in the OECD for both residential and business customers (Figure 28). Competition has also increased in mobile telephony since a second operator entered the market in 1998. This contributed to a sharp rise in the number of mobile subscribers per 100 inhabitants from 22.0 in 1998, around the OECD average, to 48.9 in 1999,[44] well above the OECD average (32.4). There are no plans to increase competition further by opening this market to a third operator owing to doubts about the technical and/or economic viability of such a move. However, these doubts should be resolved by offering a licence and seeing if there are any takers; as things stand, Luxembourg is one of only four OECD countries (the others are Iceland, New Zealand and Norway) that still has a duopoly in mobile communications and the other three are moving to increase the number of operators. In the event that competition in mobile telephony can be increased, this would provide a platform for greater competition across a spectrum of telecommunications services when third generation (3G) mobile services become available. The authorities plan to offer four licences for 3G mobile services, although again it remains to be seen whether this many licences would be technically and/or economically viable.

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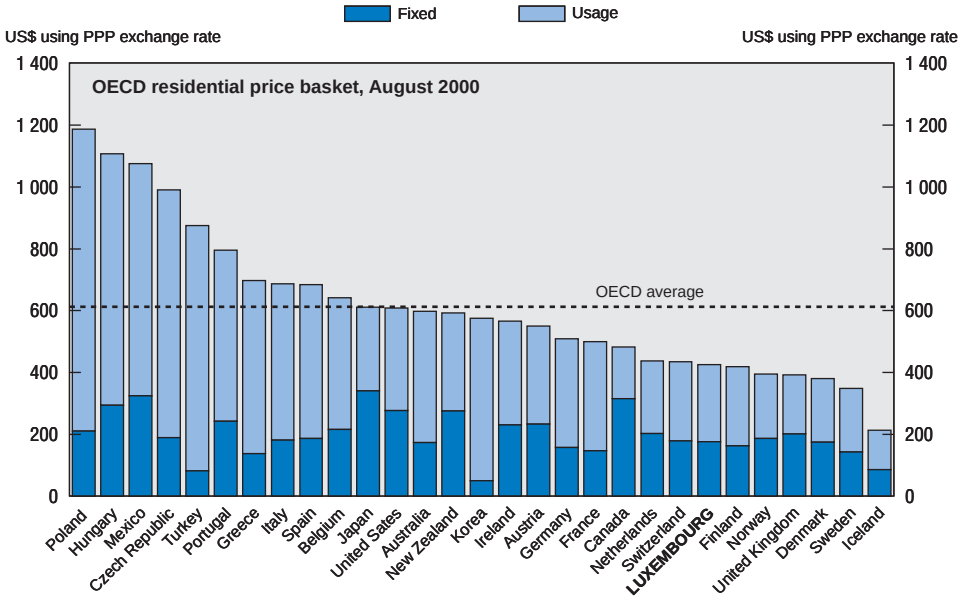
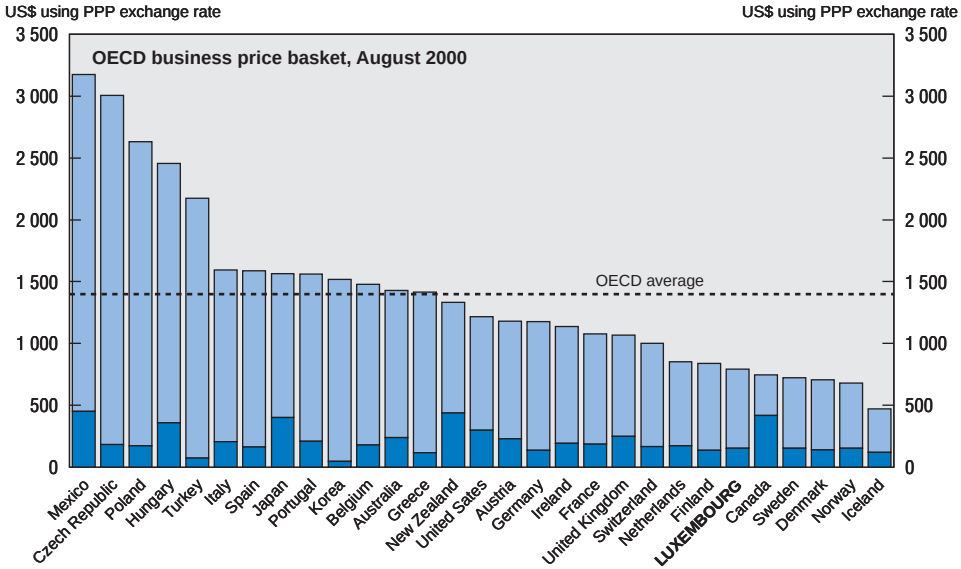
Figure 28. **Residential and business telecommunications tariffs**[1]





1. Composite basket includes international calls and calls to mobile networks.
Source: OECD, *Communications Outlook 2000*.

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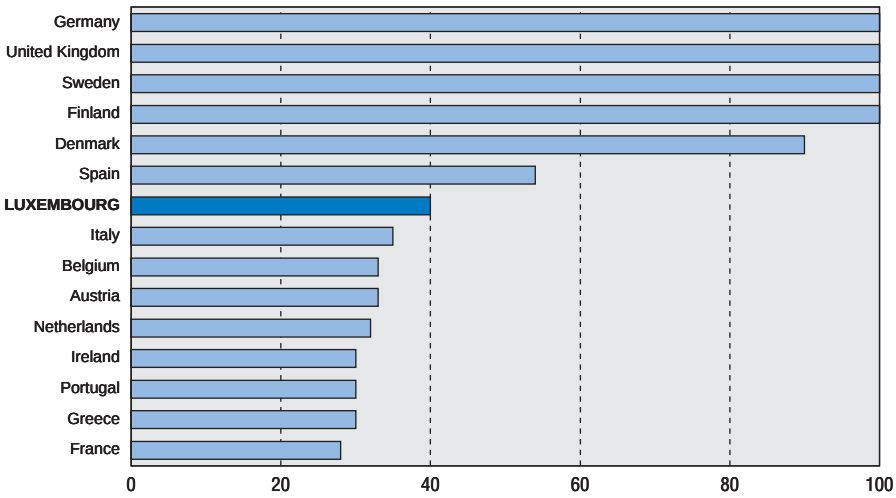
While progress has been made in opening the *electricity* market to competition, this process is less advanced than in some other EU countries (Figure 29). Some 40 per cent of the market is already open to competition and a law passed this year provides for the progressive liberalisation of the market over the next few years. In the first stage of this reform, customers buying over 20 gigawatts will have free choice of supplier from January 2001. This limit will be reduced to nine gigawatts in January 2003 and one gigawatt in January 2005. This reform should reduce electricity prices for business customers. Given that similar reforms are occurring in all EU countries, it is difficult to assess what effect it will have on electricity prices in Luxembourg relative to those in other EU countries. Electricity prices for (mainly) business customers (over 2 000 kWh) are around the EU average in PPP terms[45] while retail electricity prices (customers for 3 500 kWh in 1997) compare favourably with neighbouring countries and the EU.

Liberalisation of the *gas* market is at an even earlier stage: no liberalisation has yet occurred, but a new law that will bring Luxembourg into conformity with its Single Market commitments should be legislated next year. A Bill transposing the EU Directive 98/30/CE of 11 May 1998, concerning common rules for the natural gas market was presented to Parliament on 17 August 2000 and will be voted on as soon as constitutional procedures have been implemented. The

Figure 29. **Openness of the electricity network in 2000**



Source: European Commission.

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project provides for the opening of more than 51 per cent of the market from the beginning. This should rise to more than 70 per cent by the end of 2001 and then steadily increase to around 100 per cent in 2010 (compared with 33 per cent required by the Directive in 2008).

Regarding *postal services*, the Bill transposing the Directive 97/67/CE concerning common rules for the development of the domestic market in postal services in the EU and for the improvement in service quality was passed by Parliament on 30 November 2000. This law reserves postal services in mail weighing less than 350 grams for the universal service operator. This limit was chosen in view of the relatively large proportion of cross-border mail in Luxembourg (36 per cent of the total compared with an EU average of 4 per cent). Nevertheless, even with this limit, 7 per cent of the postal market has been opened to competition, compared with 3 per cent for the European postal market. Insofar as the *railways* are concerned, EU Directives 91/490/CEE, 95/18/CE on access to rail infrastructure and its use were transposed into law in June 1999.

Overall, Luxembourg was slow in transposing Single Market directives into law until one year ago, but has made considerable progress since then. As of November 1999, it had the second worst record, with implementation being particularly poor in the areas of transport, motor vehicles, plant-health checks and social policy (EU 1999, pp.7-8). Moreover, Luxembourg had one of the lowest rates for resolving legal disputes with the European Commission for breaches of Internal Market directives (*ibid*, p.11). In part, this record was attributable to the time-consuming consultation process required to modify the law in Luxembourg. The new government has given high priority to reducing the backlog in transposing Single Market directives into national law, as evidenced by the EU S*coreboard*, which shows Luxembourg rising to ninth place in 2000 from fourteenth place one year earlier and the rate of non-transposition of Directives into national law falling from 6.5 to 3 per cent. There are plans to adopt a new competition law, as recommended by the EU (EU, 2000*b*), but no official draft proposals have yet been introduced into the legislative process. The EU's recommendation is aimed at enabling the authorities to pursue a more active competition policy, giving them the power to enforce EC articles 81 (cartels) and 82 (abuse of dominant positions).

Developing e-commerce

Infrastructure developments[46]

An important condition for the development of electronic commerce in most countries is to have a high proportion of the population using the Internet. While internationally comparable data on users are not available, it is possible to proxy user numbers by Internet hosts,[47] as they are highly correlated with the number of Internet users where data are available. Internet host penetration in

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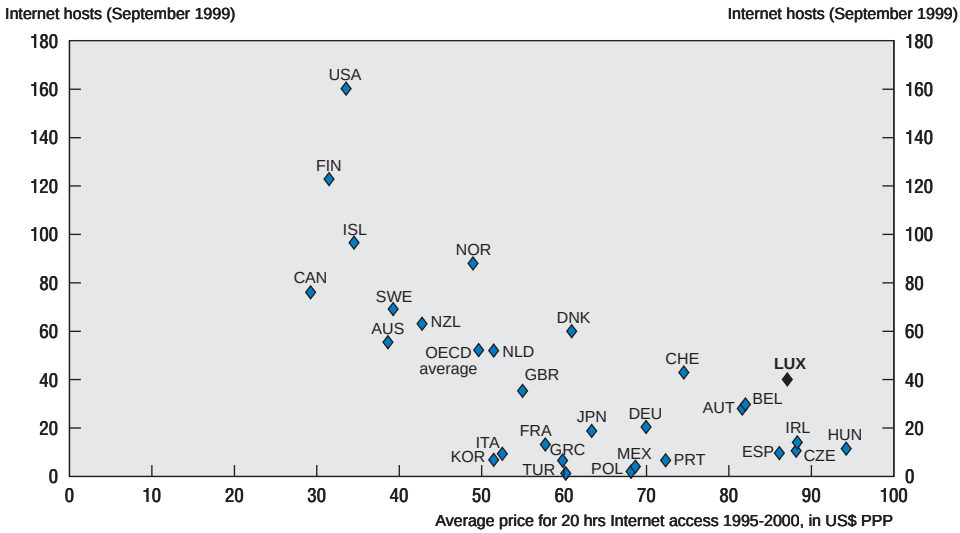
Figure 30. **Internet access costs and Internet host penetration**[1]



1. Data on hosts for Luxembourg are from mid-1999. Internet access costs include VAT.
Source: OECD, *Communications Outlook 2000*.

Luxembourg is below the OECD average, although still higher than in most other EU countries[48] (Figure 30). At the same time, the cost of Internet access has been high over the past five years. This cost is significantly correlated with the penetration of Internet hosts in OECD countries: in countries where access costs have been the highest over time, such as Luxembourg, the penetration of Internet hosts tends to be the lowest. The cost of Internet access in Luxembourg has been declining relative to the OECD average in recent years but was still very high when the last survey of charges was undertaken in September 2000 (Figure 31). However, substantial reductions in Internet access charges were announced two weeks after this survey of charges was published.

Luxembourg has a high number of secure socket layer (SSL) servers per capita, suggesting that electronic commerce has developed rapidly despite low Internet user penetration (Figure 32). The number of secure servers per capita is an important indicator of the development of electronic commerce as they are used for encrypted credit card transactions over the Internet. The main non-retail use is for subscription access to privileged information, such as research disseminated by investment banks or on-line banking. In general, countries with a high number of secure servers per capita also have a high number of Internet hosts

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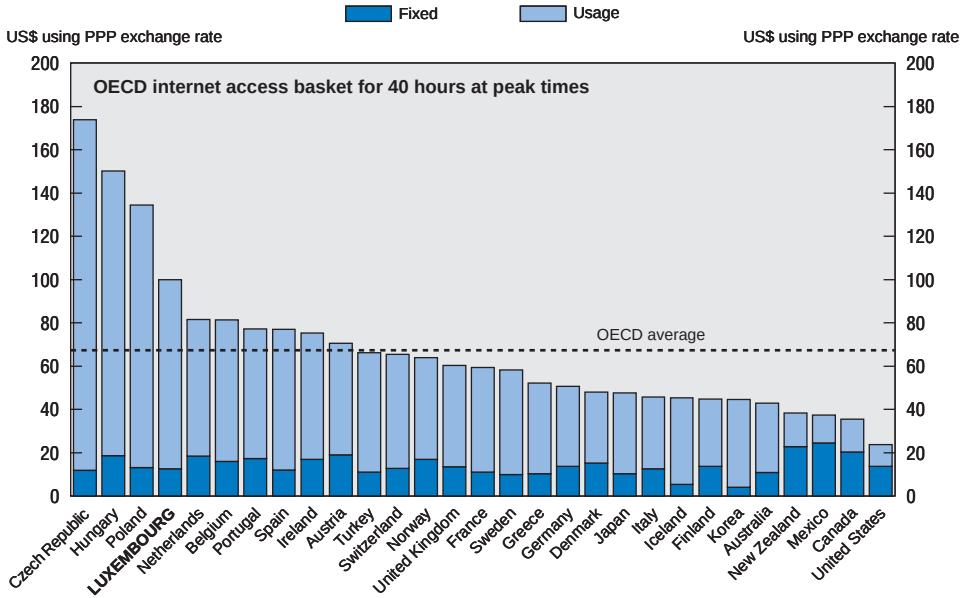
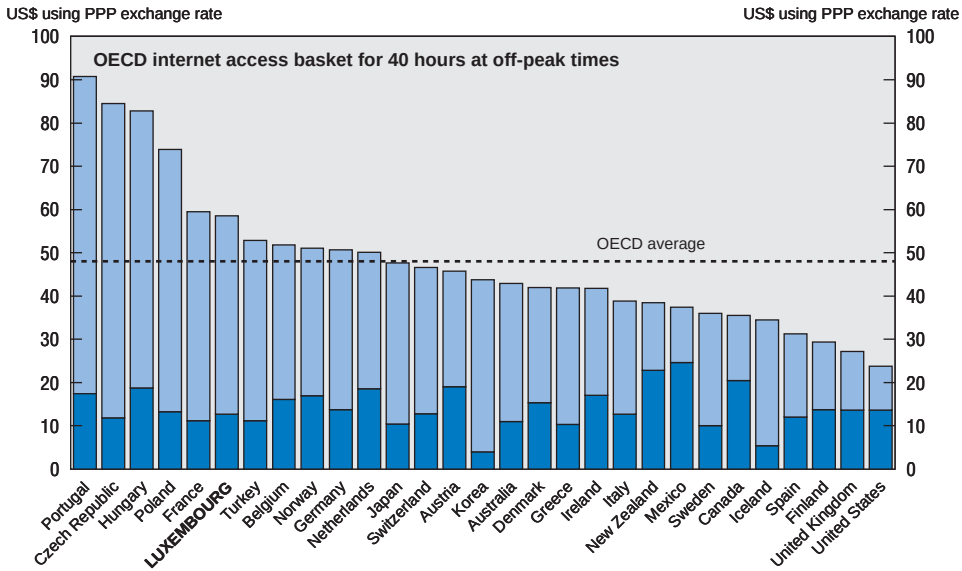
Figure 31. **Internet access charges, September 2000**[1]





1. Including VAT.
Source: OECD, *Communications Outlook 2000*.

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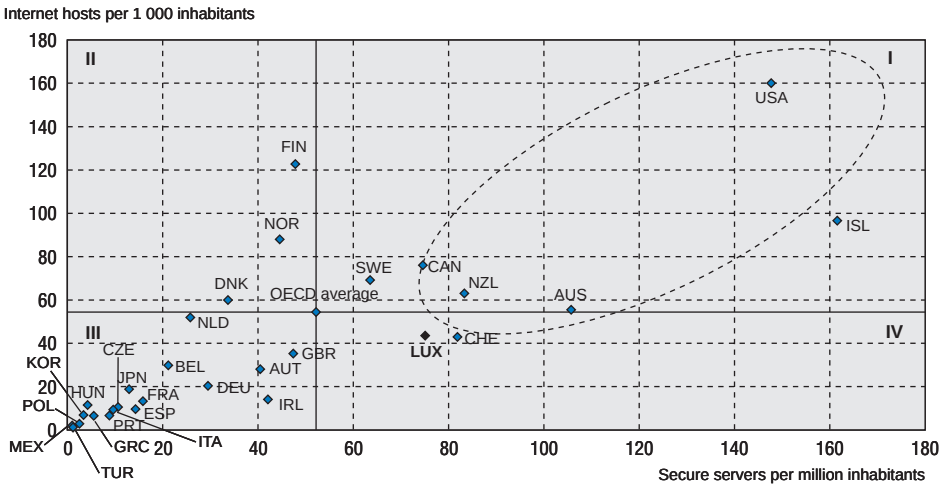
Figure 32. **Internet host penetration and secure server penetration: the role of non-metered access**[1]



1. Data on secure servers are from January 2000. Data on Internet hosts are from September 1999.
Source: OECD, *Communications Outlook 2000*.

per capita. Luxembourg and Switzerland are the only OECD countries to have an above average number of secure servers per capita but a below average number Internet hosts per capita. This undoubtedly reflects the importance and international orientation of the financial sector in these countries.

The structure of pricing, notably for consumers and small businesses, also appears to be an important determinant of the development of infrastructure for electronic commerce. Of the countries with above average host- and secure server penetration, all but Sweden and Iceland have non-metered access charges; both Sweden and Iceland have relatively inexpensive metered access charges (see Figure 32). The countries in the second quadrant of Figure 32 have low metered access charges.[49] This appears not to have hindered Internet access but to have been a brake on the development of electronic commerce. In Luxembourg and Switzerland, the countries in the fourth quadrant, metered access charges appear to have constrained Internet access but electronic commerce has nevertheless developed to a greater than average extent, probably reflecting the importance of the financial sector. Even so, electronic commerce would likely have developed to an even greater extent had there been low, non-metered access charges, as in the English-speaking countries in quadrant I.[50] The countries in the third quadrant, with below average Internet host penetration and secure server

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penetration, typically have had high, metered access charges. Some telecommunications carriers in this group – in Germany, Hungary and the United Kingdom – have announced the introduction of non-metered telecommunication access charges in 2000 to accelerate the development of electronic commerce and Internet access. The authorities have moved to encourage a similar development in Luxembourg by opening the local loop (the "last mile") to competition. Experience in other countries suggests that this is the best way of achieving low non-metered access charges and hence, of putting into effect the commitment made by EU countries at the Lisbon summit in March 2000 to take steps to reduce Internet access charges. The high proportion of households with television cable access (95 per cent) or at least a mobile telephone (62 per cent in June 2000) provides a strong basis for widening the choice of Internet connections and thereby further intensifying competition between Internet service providers.

New e-commerce legislation

A new law giving an electronic signature the same legal standing as a hand-written one was passed in July, making Luxembourg the first EU country to pass such a law. An electronic signature is a code issued by a certification service provider. For the signature to be valid, the secure arrangements for creating the signature must be under the exclusive control of the person signing, the certification service provider (which will remain under surveillance) must meet certain conditions and the data to which the signature relates must be complete. The law also establishes that electronic commerce falls within the jurisdiction of the supplier's country of origin and that intermediaries transporting, caching (*i.e.* temporary storage) or housing data have limited responsibility under certain conditions.[51] This law should help to reinforce Luxembourg's favourable starting position in the development of e-commerce, especially in relation to marketing financial services over the Internet. Separate laws being prepared on consumer protection and protection of personal information will complement this law.

The advantage of a comparatively low Value Added Tax (VAT) rate

The standard VAT rate in Luxembourg is 15 per cent, the lowest rate permitted in the EU. This makes Luxembourg a more attractive base than other EU countries for sales of goods and services *via* the Internet. If the draft EU directive (issued in June 2000) on the VAT treatment of e-commerce is approved, Luxembourg stands to gain sales and a substantial revenue windfall. The directive aims to create a level playing field in e-commerce for EU and non-EU companies. Presently, companies from outside the EU selling goods and services in the EU *via* the Internet are not subject to VAT whereas EU-based companies are subject to VAT both on sales within and beyond the EU. The directive would require non-EU companies to register in an EU country and pay VAT there on their sales *via* the Internet within the EU;

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this would also bring the tax treatment of such sales more in line with that for sales by traditional means, which are subject to VAT. EU companies would continue to be subject to VAT on sales within the EU but would be exempt for exports *via* the Internet, as they presently are for exports sold by traditional means. Many non-EU companies selling goods and services *via* the Internet in the EU could be expected to register in Luxembourg so as to pay the lowest permitted VAT rate.

Framework conditions for the diffusion of ICT[52]

Tentative evidence is emerging that the diffusion of information and communications technology (ICT) in the United States has increased multi-factor productivity (MFP) growth in recent years, mainly by increasing the quality of the capital stock.[53] Luxembourg, like most other OECD countries, is far behind the United States in the diffusion of these technologies and has not yet experienced an increase in MFP growth. This raises the issue of which framework conditions are likely to be important for the diffusion of ICT.

Exploiting the potential of ICT to raise productivity in existing firms and activities often entails significant changes in production arrangements. For example, firm hierarchies tend to be flatter with ICT because senior management has direct access to more of the information it needs without having to go through middle managers. ICT also enables firms to operate with lower inventory levels and to adapt production more quickly to changes in demand. For existing firms to be able to adapt their production arrangements to take advantage of these possibilities, flexible labour markets are required. These enable firms to transfer employees from functions that have become redundant to new functions or, if employees are not suitable, to replace them with others who have the necessary skills. As noted above, employment protection regulation in Luxembourg is strict, limiting the scope for such measures. Flexible working time is also important for enabling firms to adapt production schedules to demand, thereby minimising inventories. While Luxembourg has taken a step towards increasing working time flexibility, this is still rather limited by international comparison. Another factor that can limit diffusion of ICT is the availability of suitably skilled staff. This is unlikely to be a constraint for Luxembourg given the large catchment area from which it draws and the relatively high salaries in Luxembourg. The measures (paragraphs above) aimed at improving the education system, especially those intended to raise education attainment, as well as those aimed at increasing participation in work-related continuing education, should increase the proportion of residents occupying these jobs. Industry-research relationships and networking are also likely to be important factors influencing diffusion of ICT. These are rather weak in Luxembourg owning to the virtual absence of higher education institutions.

The pressure on management of existing firms to raise productivity is also likely to be an important determinant of the diffusion of ICT. This pressure comes

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from competition in product markets and capital markets: managers who face highly competitive product markets and pressure to deliver shareholder value have greater incentives to seek out ways of increasing productivity, including through the use of ICT. Globalisation (which is reinforced by the use of ICT), growth in e-commerce and development of the Single Market in Europe have undoubtedly increased competition in the markets in which Luxembourg firms sell their goods and services and this intensification of competition can be expected to continue. On the other hand, incumbent managers in civil law countries, such as Luxembourg, have more protection from hostile take-overs than do managers in common law countries, reducing pressure to deliver shareholder value and hence to adopt productivity-raising technologies.

Luxembourg's legal system, like those of its neighbours, may also be less well suited to the development of new ICT-based firms than systems in common law countries. Legal systems derived from French civil law, as is Luxembourg's, offer less protection to minority shareholders than is the case in common law countries.[54] This tends to result in concentrated shareholdings and probably makes it difficult for firms to raise equity finance as minority shareholders fear expropriation by managers and concentrated owners. Such ownership structures are likely to be less well suited to firms in the ICT sector as their activities tend to be subject to greater risk than those in other sectors. In these circumstances, diversified equity financing is likely to be more appropriate than debt and concentrated shareholdings. Fortunately, corporate governance in Luxembourg and other European countries with similar legal systems is evolving to give more protection to minority shareholders, albeit partly in response to pressure from pension fund managers in the United States and the United Kingdom.

Other factors likely to be important for the development of ICT-based firms are regulatory costs of creating firms and the availability of venture capital. Excessive regulation in the registration of new businesses (as well as opacity in the procedures) adds costs that can discourage entry. Luxembourg's system of regulation for new companies is less cumbersome than the systems in most other countries, favouring firm creation (Figure 33). On the other hand, Luxembourg does not have a significant venture capital industry.[55] Budding entrepreneurs must seek venture capital abroad, notably in neighbouring countries. Venture capital markets in Luxembourg's neighbours have begun to develop since 1995, but remain small compared with those in the United States.

Taxation is also likely to affect the choice of location of ICT-based firms: onerous taxation of projects that succeed would tend to encourage such firms to establish elsewhere. As noted above, Luxembourg is reforming its income tax system to ensure that it remains an attractive location for firms and for high-income individuals.

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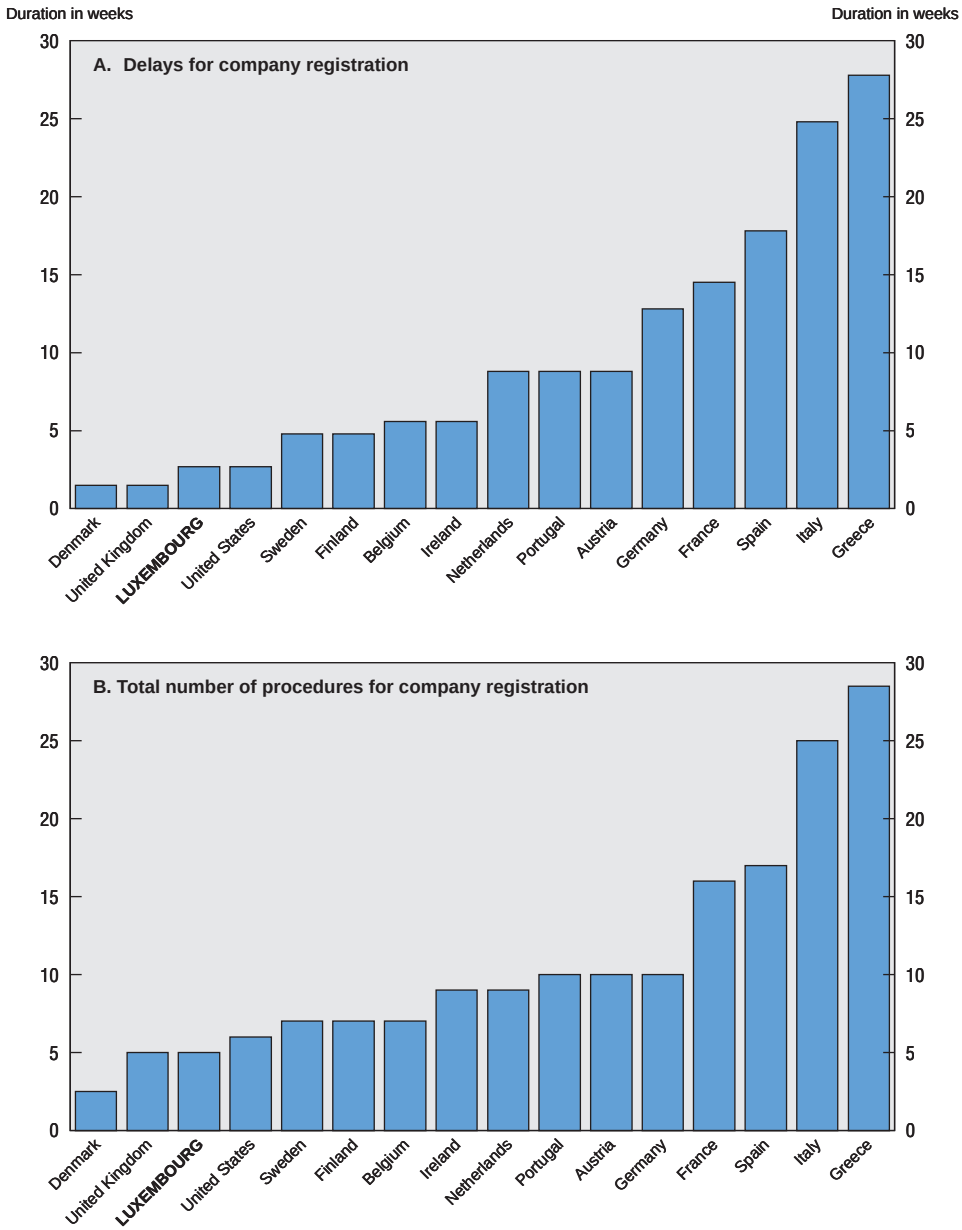
Figure 33. **Regulatory burdens for company registration**



Source: Logotech Study, 1997.

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Financial sector

The financial sector is centred on three main activities: international (private) banking; investment fund management; and insurance (life insurance and reinsurance).[56] Each of these sub-sectors is large by international comparison; indeed, the investment fund management sector is the second largest in the world, with more than € 900 billion in assets under management. All of these activities are growing rapidly. Real value added in the financial sector grew by 9 per cent per year on average during the 1990s and the sector now accounts for about 20 per cent of GDP. Throughout the period of rapid development of this sector, which dates back to the early 1980s, the authorities have placed a high priority on ensuring that the legal framework in Luxembourg remains conducive to continued development of the sector. This has entailed identifying and quickly removing any legal barriers to development in financial market niches. EU financial market directives have been implemented quickly and flexible legal frameworks have been adopted.[57] This approach has enabled the financial sector to exploit synergies between different activities and, in so doing, to diversify away from private banking. The government's proactive approach to legislation has again been in evidence recently, with new laws passed on mortgage banking, pension fund management and on electronic signatures. Moreover, as noted above, related e-commerce laws are planned on protection of the consumer and protection of private information. These new laws should facilitate further development in mortgage banking, fund management and the provision of financial services *via* the Internet.

International co-operation in tax matters

Bank secrecy laws and lack of effective exchange of information for tax purposes in a number of countries, including Luxembourg, have prevented foreign tax authorities from obtaining information on their residents' income on savings held generally with financial institutions in these countries. For non-residents wishing to evade tax liabilities in their home countries, this has made these countries a more attractive destination for financial investments than other countries. The removal of exchange controls by OECD countries and, more recently, technological advances that have made international banking more accessible to a wide range of taxpayers, have considerably increased the scope for non-residents to evade taxes in their home countries in this way. In view of the large and growing scale of this kind of tax enforcement problem, all OECD countries, including Luxembourg, agreed in March 2000 to work towards improving access to bank information for tax purposes (OECD, 2000*a*).

It was recognised that achieving this improved level of access would require more substantial revisions to laws and practices in some countries than others. Accordingly, the OECD's Committee on Fiscal Affairs encouraged Member countries to take a number of incremental steps towards this goal. Luxembourg's

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Parliament has indeed already taken a step in this direction by finalising the ratification of the additional Protocol on tax matters to the European Convention in Mutual Assistance in Criminal Matters in October 2000. Signatories to the Convention that have also ratified the additional Protocol will now be able to request information, including bank information, from Luxembourg for the investigation and prosecution of criminal tax matters. Parliament also ratified the US Mutual Legal Assistance Treaty with the United States in October 2000. This treaty provides for exchange of information in criminal cases, including fraud cases.

Moreover, it is to be noted that the EU Council agreed at the Feira Summit in June 2000 on the key elements of a future Directive on the taxation of savings. This Directive, which will apply only to non-resident individuals of another member state, aims to establish after the end of the transition period a system of information exchange to ensure a minimum effective taxation of savings income in the form of interest payments within the EU. This will entail setting up a network that will provide information on interest payments to the relevant tax authorities. Implementation of the different stages is conditional on equivalent measures being taken in the United States and key third countries (Switzerland, Liechtenstein, Monaco, Andorra and San Marino) and on the same measures being taken in all relevant dependent or associated territories (the Channel Islands, Isle of Man and those in the Caribbean). Once sufficient reassurances have been obtained with regard to the application of equivalent measures in the named countries and the same measures in the dependent or associated territories, the EU Council will decide by unanimity on the adoption and implementation of the Directive. This is to occur no later than 31 December 2002. The ECOFIN Council at its meeting of 26 November 2000 recalled the conditions governing transition from each stage to the next as set out in its Report to the European Council at Feira and its annex and approved the essential content of the future directive on the taxation of savings income. Luxembourg, Belgium and Austria will operate a withholding tax instead of participating in the information exchange network until the end of the transition period. The minimum withholding tax rate to be applied by these countries will be 15 per cent for the first three years of the transitional period and 20 per cent for the remainder of that period. Member States that operate a withholding tax will transfer 75 per cent of their revenue from the tax to the investor's state of residence.

Luxembourg is also participating in the OECD project to identify and eliminate harmful tax practices involving geographically mobile financial and other service activities.[58] A number of preferential tax regimes have been identified in Luxembourg as potentially harmful.[59] These, along with those in other Member countries, will be examined to determine whether they are actually harmful and, in the event that they are, how the harmful features could be removed. As Luxembourg, together with Switzerland, abstained from the initial (1998) Report (OECD, 1998*b*), it reserves the right not to remove harmful features of its preferential tax regimes.

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Notes

1. For a more detailed account of economic developments in 1999, see STATEC (2000*a*).
2. A new insurance for the care of (mainly elderly) dependent persons (*l'assurance dépendance*) came into effect on 1 January 1999. As it provides benefits in kind, it is classified as public consumption under ESA95 rules. This insurance amounted to LUF 4.6 billion of public consumption expenditure in 1999 and LUF 6.1 billion in 2000. The resulting increase in GDP is somewhat less than indicated by these figures because some of this expenditure may have been classified as private consumption beforehand.
3. Aircraft purchases rose from LUF 7.3 billion in 1998 to LUF 27.6 billion in 1999. Gross fixed investment rose from LUF 127.7 billion to LUF 164.0 billion over the same period.
4. Underlying inflation excludes petroleum products, solid fuels, coffee, tea, cocoa, potatoes and cut flowers.
5. This wage agreement provides for a 3.5 per cent salary increase backdated to January 2000, which is in addition to index-linked increases, annual instead of biannual seniority-based salary rises, increases in meal and travel allowances and an extra day of annual leave per year. This will contribute around 0.5 percentage point to average growth in wage rates in 2000 (Economist Intelligence Unit 2000, p. 43).
6. Export goods prices fell by 3 per cent and import goods prices declined by 1.8 per cent.
7. Goods exports were 15 per cent higher in the first nine months of 2000 than one year earlier. This increase in export growth is mainly attributable to the steel industry, which increased exports by 22 per cent.
8. These projections were finalised on 7 November 2000. Economic data that became available since this date suggest that growth could be a little weaker than projected, notably in 2001. As a matter of fact, taking into account the most recent data, STATEC is expecting a growth rate of 5-5½ per cent in 2001. As usual, developments in the financial sector, which accounts for some 20 per cent of GDP, could heavily influence the final outcome, and indeed, could even compensate for the loss of momentum in the more cyclical sectors.
9. Due to the statistical changeover from ESA79 to ESA95, figures for the surpluses recorded over the past few years were also considerably higher than foreseen in the Stability Programmes submitted by Luxembourg to the European Commission up until early 2000.
10. The figures mentioned here are partly based on an analysis by STATEC (2000*b*).
11. It provides effective replacement rates of 90 per cent, compared with an OECD average of 60 per cent (Blöndal and Scarpetta, 1990).
12. IMF (2000*b*), "Stress Testing Luxembourg's Public Pension System", pp. 15-26.

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13. A discount rate of 6 per cent was used, but even a somewhat higher discount rate would not change the picture very much.
14. This covers sectors in which outputs are valued in the market; it includes imputed rentals on owner-occupied housing. Banks are excluded owing to the difficulty of evaluating growth in the volume of their services. The non-bank market sector amounted to 77 per cent of total value added in 1998.
15. Productivity cycles have been determined by identifying productivity peaks. These occur in years in which the gap between actual and trend MFP growth turns from increasing to decreasing. Trend MFP has been calculated using a Henderson 11-period moving average (see Figure 11). This methodology for determining productivity cycles is taken from the Australian Bureau of Statistics.
16. The 1995-99 productivity cycle is not yet complete.
17. Blanchard and Wolfers (1999) develop a model to explain the role of shocks and institutions in the rise in European unemployment. They are able to explain both the evolution and the cross-country patterns of unemployment. The adverse shocks that contributed to the rise in unemployment were; a decline in multi-factor productivity (MFP) growth; a rise in real interest rates; and adverse labour demand shifts (unrelated to movements in real product wages). These shocks interacted with labour market institutions that made increases in unemployment long lasting. The most significant such institutions are high benefit replacement rates and long benefit duration.
18. Estimation of impulse responses from a simple, unrestricted two variable VAR-model indicates that cross-border employment responds significantly more strongly to an impulse in domestic employment than does national employment.
19. Specifically, a disability pension may be granted to someone who loses their capacity to work such that they cannot exercise another occupation that corresponds to their strengths and aptitudes in relation to the general labour market. To qualify for such a pension, the person must have contributed to social security for at least one year during the three years before the onset of the disability. These three years must also post-date the claimant's 25th birthday.
20. Contributions may be imputed for study undertaken between the ages of 18 and 27, rearing children aged less than 6 or caring for a dependent person. In such cases, the imputed years and 15 years of actual contributions must pre-date the creation of pension arrangements for persons not receiving a salary.
21. This arrangement was created in response to unemployment during the crisis in the steel industry and was subsequently adopted by some other sectors.
22. The estimates for some of these groups are not statistically significant as they are based on a small number of observations. Sample size declines with age and by level of education attainment and is smaller for women than for men.
23. Since December 1990, the law on early retirement (*préretraite*) has offered an early retirement indemnity to full-time employees aged at least 57 who agree to have their position converted into a part time position. The part-time position must be for 40-60 per cent of the time worked in a full-time position. The government employment fund refunds the employer entirely for the early retirement indemnity provided that he hires an unemployment person on a fixed term contract or an apprenticeship contract for at least the proportion of the full-time position made vacant. The law of 12 February 1999 simplified the application of this legislation by:
    - increasing the periods during which the compensating hiring could be made to six months before and after the departure on progressive early retirement;

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– reducing the period during which the additional hiring must be maintained to two years; and

– allowing for the additional hiring to be full-time, not just part-time or by way of an apprenticeship, as before.

24. The government employment fund pays the employer's share of social security contributions for an employee aged over 49 who agrees to have his full-time post converted into a part-time post provided that the employer hires an unemployed person for the part of the post freed for at least 18 months. In the case where the unemployed person hired has an indefinite term contract, all social security contributions for the person taking a progressive shift to part-time work are paid by the employment fund. The same applies where the unemployed person is hired full time or where they are a member of the under-represented sex in the sector or profession concerned. The government employment fund pays for the employer's social security contributions for the hired person for up to seven years. Where the person hired has an indefinite-term contract, a full-time past or is a member of the under represented sex in the sector or profession concerned, the fund pays for all the social security contributions due.

25. These arrangements provide an important conduit for reducing the labour force in the steel industry, which remains above required levels owing to the change from the blast furnace to the EAF method of making steel. Accordingly, *l'*ARBED, the steel making company, considers that Luxembourg's model of early retirement at the age of 57 and government support for jobs being carried out with excess labour will need to continue until the end of 2001, as decided by tripartite arrangements.

26. OECD (1994), pp. 97-98. See also J. Hunt (1999).

27. Employment protection regulation (EPR) sets rules on hiring (*e.g.* rules favouring disadvantaged groups, conditions for using temporary or fixed-term contracts, training requirements) and firing (*e.g.* redundancy procedures, mandated pre-notification periods and severance payments, special requirements for collective dismissals and short-time work schemes) (OECD, 1999).

28. There is also tentative evidence on a cross-county basis (Bassini, Scarpetta and Visco, 2000, pp. 28-32) that strict EPR, as in Luxembourg, reduces multi-factor productivity (MFP) growth. Such regulation may make firms very cautious in adjusting their labour force and may reduce returns from investment and mergers and take-overs. This would slow the pace at which resources flow to higher productivity uses, reducing growth in MFP.

29. This proportion rise to 97 per cent for national employment (*i.e.* excluding cross-border workers).

30. Working time of 30 hours or less per week and 130 hours or less per month is defined as part-time.

31. A large impact of parenthood or the employment gender gaps is not a problem if it reflects individual preferences exercised in the context of appropriate prices and opportunities. For example, the large impact in Luxembourg could reflect traditional values and high household incomes with only one partner working. On the other hand, if the large impact reflected an inadequate provision of childcare facilities, a policy response to remove this barrier to female employment would be appropriate. The government considers that inadequate provision of childcare facilities is a factor contributing to the large gender gap in the employment ratio and is taking measures to remove this barrier to female employment (see below).

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32. These training programmes include: household services; information technology trainer; and creating a small business or helping with the management of a small business.
33. These subsidies are paid under the programme known as "*action promotionelle des actions positives*". Such re-organisation of work arrangements is one of the criteria for subsidies to be paid under this programme.
34. This arrangement applies to the situation where a full-time employee moves to part-time employment within the same enterprise and a woman who has been unemployed for at least 3 months is hired. In such a case, the employers' social security contributions for that woman will be paid by the state for up to 7 years.
35. Parental leave came into effect in January 1999. It gives parents the right to six months paid leave in the year following the birth of a child. This leave is paid for by public funds.
36. Unless excluded by the employment contract, part-time workers may work up to 20 per cent longer than the daily or weekly duration of work provided for in the contract without being paid overtime rates. This possibility is subject to the condition that the average weekly working week over four consecutive weeks does not exceed the normal working week fixed in the contract.
37. This paragraph is based on EU (2000*a*), p. 143-45.
38. These issues are further developed in IMF (2000*a*), "Why is Luxembourg growing so fast", pp. 6-14.
39. Since the beginning of 2000, the index basket used excludes consumption by non-residents.
40. Minimum wage rates are 80 per cent of the adult rate for 17 year olds, 70 per cent for 16 year olds and 50 per cent for 15 year olds.
41. The corresponding proportions are 11 per cent in France, 3.8 per cent in Hungary, 17.6 per cent in Mexico, 3.7 per cent in the Netherlands, 4.5 per cent in Poland, 4.7 per cent in Portugal and 5.1 per cent in the United States (OECD 1998*a*, p. 39).
42. This paragraph draws extensively on OECD (1998*a*), "Making the most out of the minimum: statutory minimum wages, employment and poverty".
43. The information on unemployment benefit and social assistance replacement rates comes from IMF (2000*a*), p. 31.
44. This ratio rose further in 2000 to 70.2.
45. EU (1999), p.27.
46. This section draws heavily on DSTI/ICCP/TISP(2000)1/FINAL, Local access pricing and e-commerce. This document can be found at *www.oecd.org/dsti/sti/it*.
47. A host is a domain name that has an IP address "record" associated with it. This would include any computer system connected to the Internet (*via* full- or part-time, direct or dial up connections). A host may be country specific, meaning that it terminates in a country code (*oecd.fr* or *oecd.au*) or generic (such as *oecd.com* or *oecd.org* or *oecd.net*). The generic domain names are allocated to countries on the basis of registrations or on a sample of Internet Provider (IP) addresses. The latter method of allocating generic domain names to countries is not possible for Luxembourg owing to the lack of data.
48. The proportion of households connected to the Internet (36 per cent in October 2000) is also higher than the EU average (28 per cent) (Eurobarometer Survey, October 2000)

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although it remains well below the connection rates in the Netherlands, Sweden and Denmark, which are over 50 per cent.

49. The incumbent operators in Korea, Portugal and Spain introduced non-metered Internet access *via* the PSTN during 2000.

50. The incumbent operator in Luxembourg announced in October 2000, an offer of non-metered Internet access but only for ISDN connectors and only on weekends. This option is called "Flat Rate Weekend".

51. For the simple transporting or caching of data, the service provider must only play a passive role to avoid responsibility. For housing the data, the service provider is not responsible if he is not aware of illegal content or as soon as he becomes aware of such content acts to withdraw it or make it inaccessible.

52. This section draws extensively on Bassanini, Andrea, Stefano Scarpetta and Ignazio Visco (2000).

53. In other words, such technical progress has mainly been capital augmenting, not disembodied.

54. La Porta, Rafael, Florencio Lopez-de-Silanes, Andrei Schleifer, and Robert W. Vishny (1998) examine legal rules covering protection of corporate shareholders and creditors. They conclude that common law countries generally offer most protection, the French civil law countries the least with the German and Scandinavian civil law countries located in the middle. Concentration of ownership of shares is negatively related to investor protection, consistent with the hypothesis that small, diversified shareholdings are likely to be unimportant in countries that fail to protect their rights. They also note that concentrated ownership probably makes it difficult for firms to raise equity finance since minority investors fear expropriation by managers and concentrated owners. Luxembourg was not included in this study but it has legal system derived from French civil law, as do most of its neighbours (France, Belgium and the Netherlands).

55. Venture capital markets combine the financing, management and nurturing of risky projects.

56. IMF (2000*b*), p. 27.

57. I*bid*.

58. This project is described in more detail in OECD (2000*b*).

59. These are:

- Insurance – Provisions for Fluctuations in Re-insurance Companies;
- Finance and Leasing – Finance Branch;
- Fund Managers – Management Companies [taxation of management companies that manage only one mutual fund (1 929 holdings)].
- Holding company regimes and similar preferential tax regimes are also being examined in Luxembourg, as well as in Austria, Belgium, Denmark, France, Germany, Greece, Iceland, Ireland, Netherlands, Portugal, Spain, and Switzerland (OECD 2000*b*, pp. 62-67).

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Glossary

CES	Economic and Social Council (*Conseil économique et social*)
e-commerce	Electronic Commerce
EPR	Employment Protection Regulation
GNI	Gross National Income
ICT	Information and Communications Technology
IMF	International Monetary Fund
IP	Internet Provider
ISDN	Integrated Services Digital Network
MFP	Multi-factor Productivity
PPP	Purchasing Power Parity
NAP	National Action Plan to Promote Employment
PSTN	Public Switched Telecommunication Network
RMG	Minimum social income (*Revenu Minimum Garanti*)
STATEC	*Service Général de la Statistique et des Études Économiques*
VAT	Value Added Tax

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